# intro

Thursday, February 19, 2026 11:03 AM


# Segura® Platform Core Products

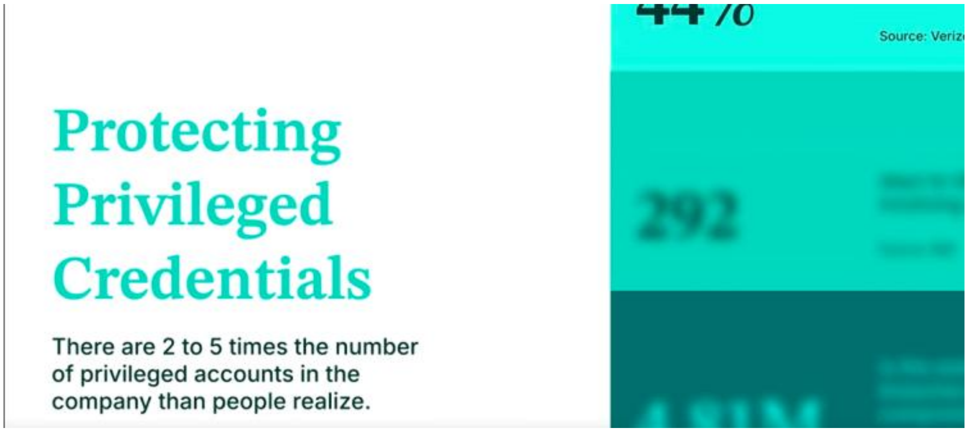
| | | |
|---|---|---|
| PAM Core | Secures, controls, and audits privileged access and critical credentials. |
| Domum Remote Access | Enables secure remote access without VPN, with full session auditing. |
| DevOps Secret Manager (DSM) | Manages and protects secrets and credentials across DevOps pipelines. |
| Certificate Manager (CLM) | Automates the full lifecycle of digital certificates (issuance, renewal, etc.). |
| Endpoint Privilege Manager (EPM) | Enforces least privilege on workstations and servers (Windows, Linux, macOS). |
| MySafe | Corporate password vault for secure storage, generation, and autofill of credentials. |
| Cloud Entitlements (CIEM) | Ensures visibility and control over identities and permissions across cloud providers (AWS, Azure, GCP). |

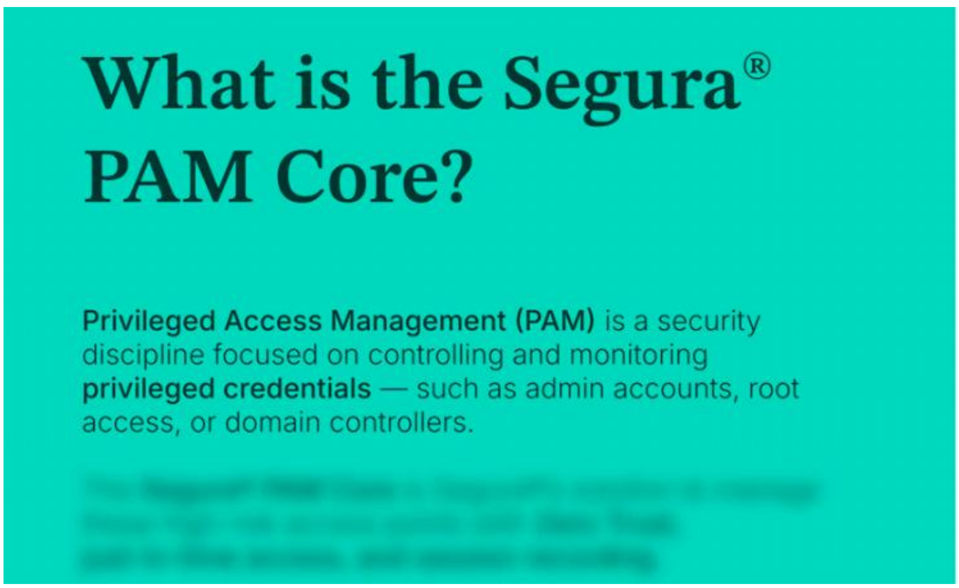
## What Other Vendors Don't Offer

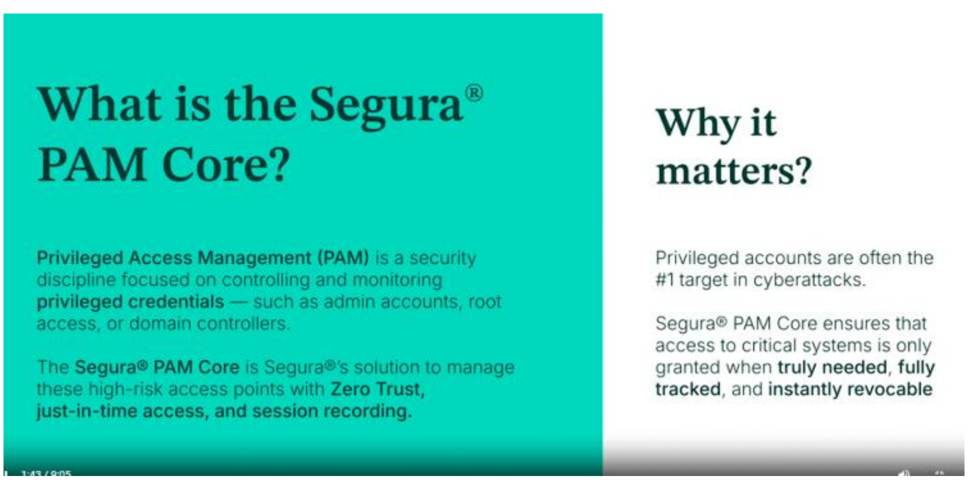
| | Segura® Platform | Traditional Vendors (e.g. CyberArk, BeyondTrust, Delinea) |
|---|---|---|
| Unified architecture across all modules | ✔ Yes | X No – multiple engines, portals, and agents |
| Integrated DevSecOps secrets + PAM + CIEM | ✔ Yes | X Usually separate or require third-party tools |
| Seamless cross-product workflows (e.g. PAM + Remote Access) | ✔ Yes | X Requires complex integrations or multiple UIs |
| One license model (TCO simplicity) | ✔ Yes | X Often separate SKUs for each module |
| All features built in-house (not acquired) | ✔ Yes | X Many core features are results of acquisitions |
| Zero Trust across the full stack | ✔ Native | X Often limited to PAM or VPN replacements |

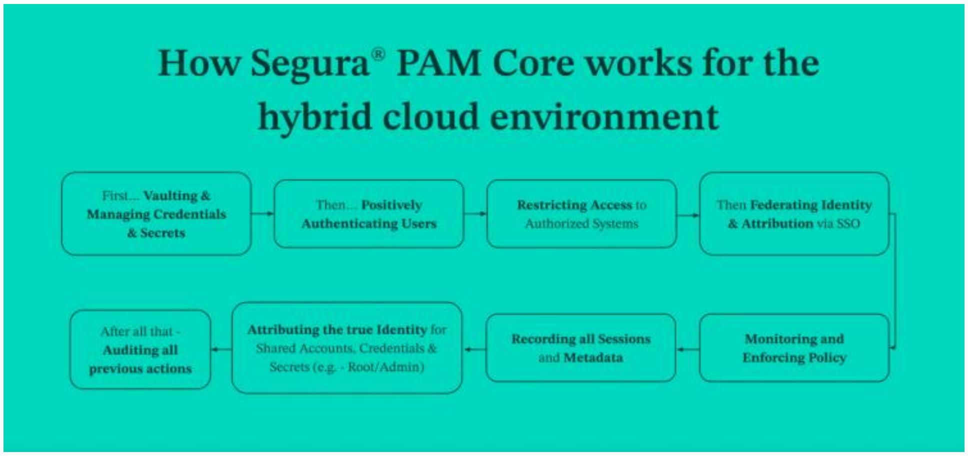
# PAM Core - concepts

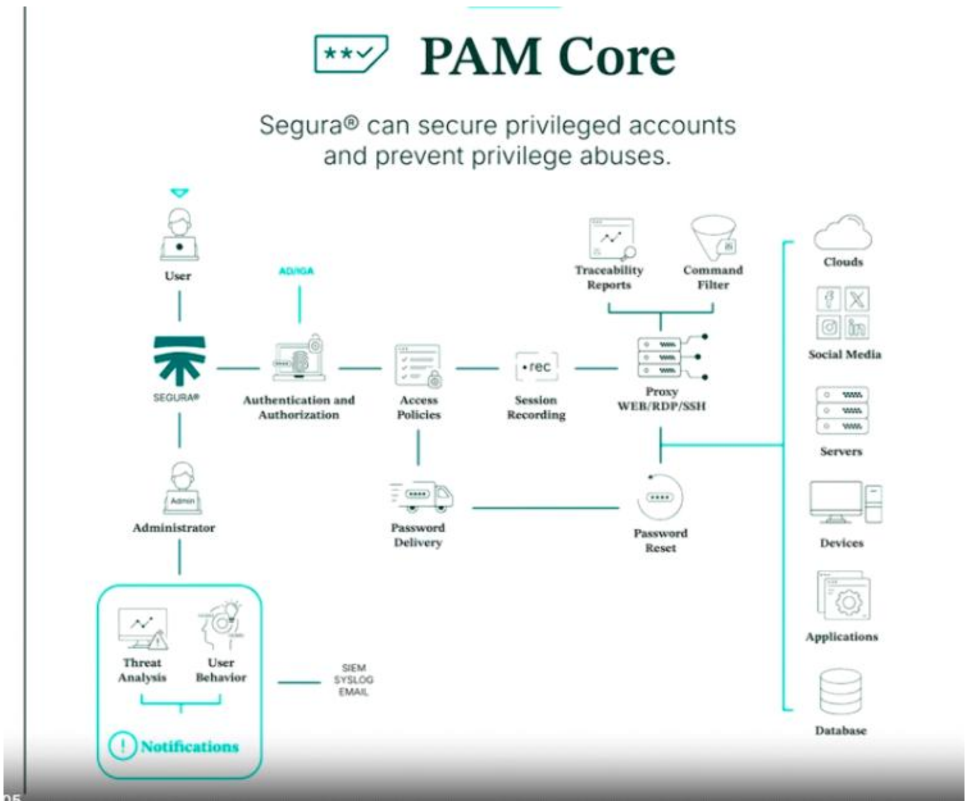
Thursday, February 19, 2026 3:32 PM






## Key Features of Segura® PAM Core

| Feature | Value |
|---|---|
| Session Recording (Video & Commands) | Enables post-incident analysis and audit trails |
| Credential Rotation | Eliminates static passwords and shared secrets |
| Approval Workflows | Access only after validation (by person or policy) |
| Keystroke & Command Monitoring | Detects suspicious behavior during sessions |
| Database Proxy Access | Manages access to DBs with full traceability |
| App-to-App Secrets Access (A2A) | [illegible] |

| | |
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| App-to-App Secrets Access (A2A) | [illegible] |

Everything organizations need to control privileged access

## Natively Integrated With Other Segura® Products

**With Domum**: Extend privileged control to remote vendors without VPN

**With DSM**: Protect secrets used in automation pipelines

**With CLM**: Uses certificates for system authentication; ensures secure machine-to-machine trust

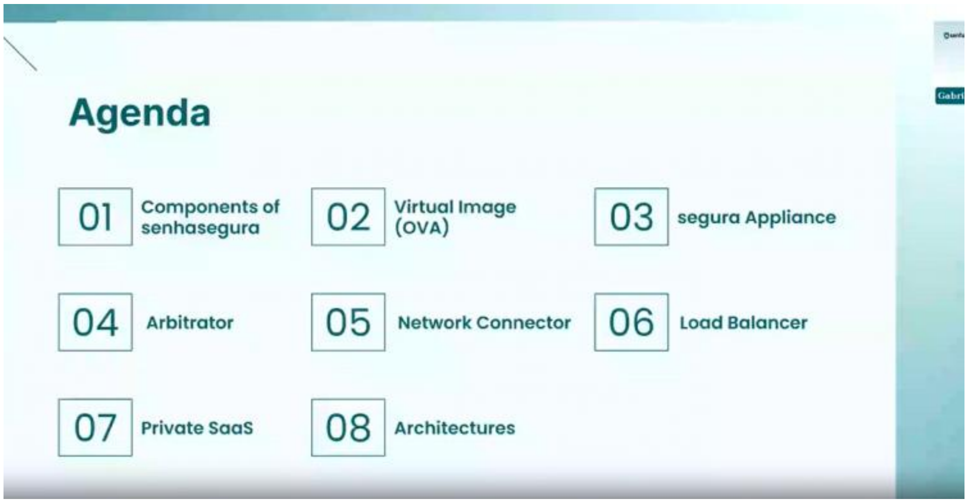
**With EPM**: Extends PAM policies to endpoints, enforcing least privilege on desktops and servers

**With CIEM**: Enforce least privilege across multi-cloud environments

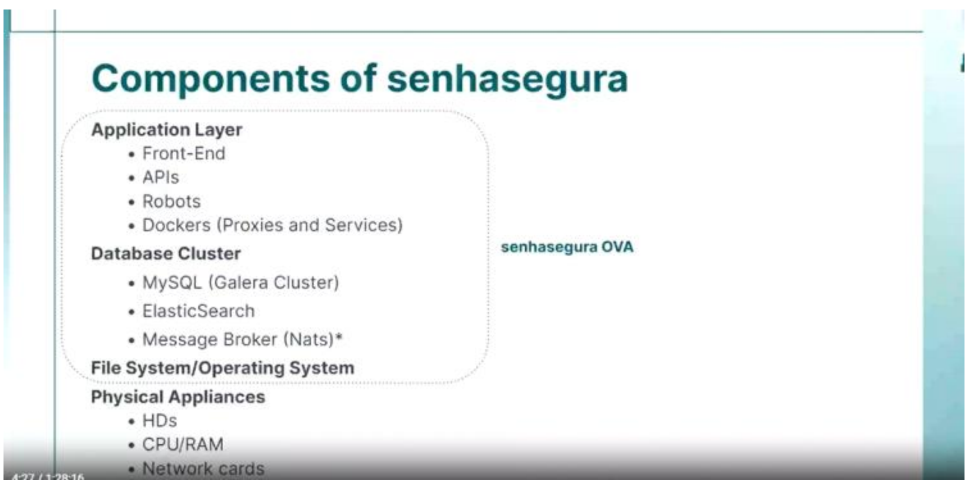
**With MySafe**: Separate personal password storage from privileged access

# Deployment & update

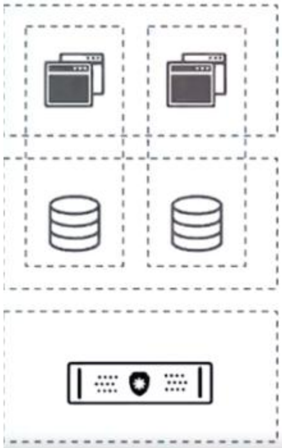
Friday, February 20, 2026 11:29 AM




## Components of senhasegura



# Components of senhasegura



**Front-End / APIs / Robots / Dockers**

- Active-Active
- Active-Passive

**Database Cluster**

- Synchronous
- Multi-Master

**Physical Cluster**

- segura Appliance
- Customer infrastructure with virtualization (**On Premise** or **Cloud** or **Híbrido**)

# Virtual Image (OVA)

## Virtual Image (OVA)

In order to facilitate the implementation of senhasegura, the main way of distributing the application is through Virtual Images (OVA).

As it is an All-in-One application, this format delivers all the components of the solution in a single artifact.

**Feature Prerequisites for Version 3.32**

| Item | Requirement |
|---|---|
| **vCPUs** | 8 |
| **Memory** | 16 GB |
| **Hard Drive** | 250 GB |
| **Network** | Gigabit 1000 Mbps |

# Virtual Image (OVA)

**Virtual Appliance formats available**

- OVA
- VMDK
- RAW
- VHD
- QCOW2

**Important info**

senhasegura does not offer Virtual Appliances in ISO file format.

**Supported Cloud Service Providers**

- AWS
- GCP
- Azure

# 03

## segura Appliance

## segura Appliance

**Introducing the segura Appliance** (previously known as PAM Crypto Appliance)

Experience the ultimate solution in privileged access management with the secure Appliance.

Our wide range of Appliances - **STEEL**, **TITANIUM** and **DIAMOND** - is designed to meet the diverse needs of organizations, offering unparalleled security, scalability and reliability.
Deployed on-site in your data center, the secure Appliance guarantees regulatory compliance and optimal protection for your critical assets.

Elevate your organization's security posture and cyber sovereignty with the segura Appliance.

It is basically a **pre-built physical server appliance** used to run the **senhasegura Privileged Access Management (PAM) platform** in **large enterprise environments**.

# STEEL

Perfect for small and medium-sized organizations, the **STEEL** model supports up to 600 remote sessions and offers 360,000 hours of recording time.

1 x Intel Xeon Gold 5315Y 3.2G, 8C/16T, 11.2GT/s, 12M Cache, Turbo, HT
1 x 32GB RDIMM, 3200MT/s, Dual Rank, BCC
2 x 12TB 7.2K RPM NLSAS 12Gbps 512n 3.5in Hot-Plug Hard Drive, configured in RAID 1

# TITANIUM

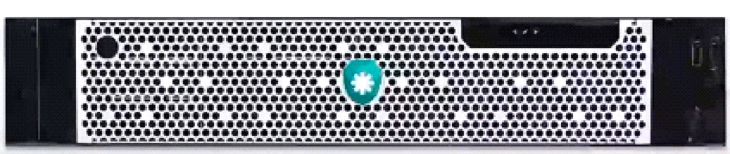
For larger organizations, the **TITANIUM** model supports up to 1,500 remote sessions and offers 720,000 hours of recording.

2 x Intel Xeon Gold 5315Y 3.2G, 8C/16T, 11.2GT/s, 12M Cache, Turbo, HT
2 x 32GB RDIMM, 3200MT/s, Dual Rank, BCC
3 x 12TB 7.2K RPM NLSAS 12Gbps 512n 3.5in Hot-Plug Hard Drive, configured in RAID 5

# DIAMOND

The premium choice for Enterprise-level organizations, the **DIAMOND** model supports up to 2,000 remote sessions and provides an impressive 1,440,000 hours of recording.

2 x Intel Xeon Gold 6346Y 3.1G, 16C/32T, 11.2GT/s, 36M Cache, Turbo, HT
4 x 32GB RDIMM, 3200MT/s, Dual Rank, BCC
4 x 16TB 7.2K RPM NLSAS 12Gbps 512n 3.5in Hot-Plug Hard Drive, configured in RAID 5



04

## Arbitrator senhasegura

## Network Connector senhasegura

The **Network Connector** is a senhasegura module that allows users to carry out sessions on devices located on networks that do not have connectivity to senhasegura, or on networks with overlapping IPs.

In addition, the Network Connector supports all types of connections available on senhasegura, such as **RDP, SSH, Telnet, HTTP/HTTPS**, among others.

This solution consists of the Network Connector Server and the Agent.

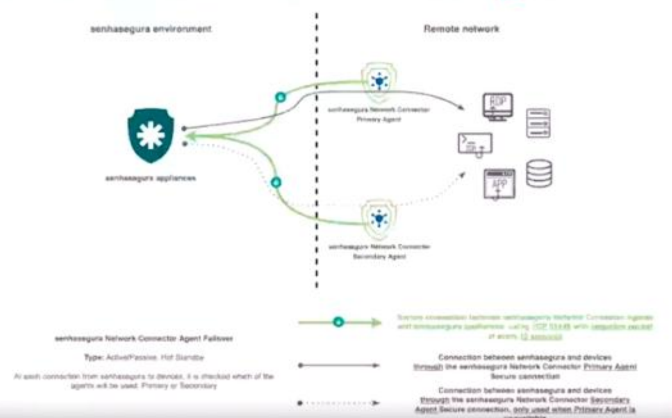
The **high availability** of the senhasegura **Network Connector** is a feature of the product, which works in Active-Passive mode. All data traffic is directed to the primary senhasegura and in the event of its failure, the secondary senhasegura comes into play and all data traffic is redirected to it.

**Important**

- The device that has the agent installed must be able to see senhasegura for the connection to be made.

- By changing the request destination to a tunnel established by the Agent, the Network Connector becomes incompatible with the use of certificates. This is because, in this context, certificates depend on the request destination to perform one of the necessary validations.

## Network Connector senhasegura



# Load Balancer senhasegura

senhasegura has a **load balancer**, developed by the same senhasegura team, to interface all the senh
instances that are in a cluster scheme. The senhasegura Load Balancer is a ready-to-deploy solution a
a ready-to-use solution, configured to give the best performance on senhasegura's functionalities.

**Technical information**

Minimum requirements

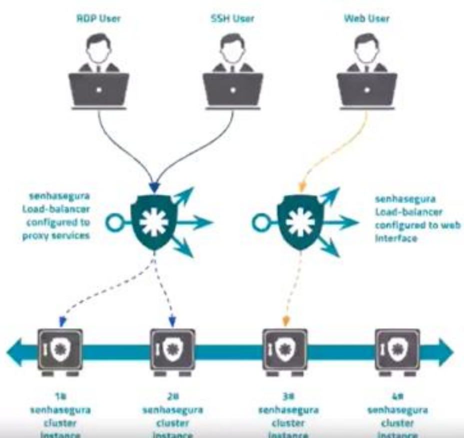
- 4GB RAM
- 2 vCPUs (4 vCPUs recommended)
- 40GB vHD

It will use the same SSL certificate installed on the senhasegura instances. There is no need to install a
specific SSL certificate for the load balancer.

As a product tailored to balancing requests to senhasegura, senhasegura Load Balancer* has a very s
optimization and should not be used to balance other applications.

Therefore, we do not offer the option of changing the balancing method and/or monitoring methods fo
client. Its use is via the **least connection** algorithm for the balancing mode.

# .oad Balancer senhasegura




# 07

## Private SaaS

# Private SaaS

**Private SaaS** is a service delivery approach that eliminates the need for server or controller installations, resulting in cost savings, reduced administrative demands and the elimination of on-site software or hardware maintenance and upgrades.

**Key features**

**senhasegura cloud on Google Cloud (GCP):** our service operates on Google Cloud infrastructure, guaranteeing reliability and scalability.

**Virtual Private Cloud for Each Client:** we offer a dedicated virtual private cloud for each client, guaranteeing total data segregation. Total segregation protects against configuration errors, data leaks or any adverse impact of one customer's activities on another.

**Compliance with Safe Harbor and Data Protection laws:** our platform is designed to adhere to strict Safe Harbor laws and data protection regulations, guaranteeing the security and privacy of your data.

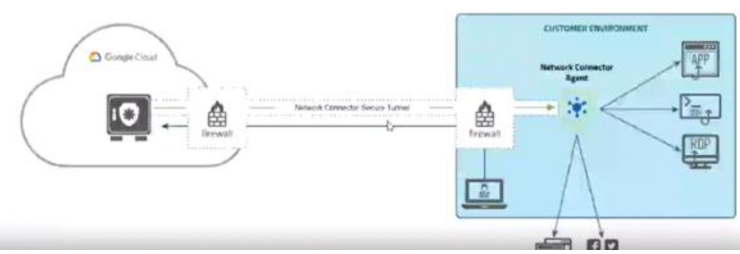
**99.9% SLA (Service Level Agreement) per year**: we commit to an uptime SLA of 99.9%, guaranteeing uninterrupted access to our services. Please note that scheduled maintenance is not included in this SLA. For more details, visit the Compute Engine Service Level Agreement provided by Google Cloud.

# Private SaaS - Architectures

**Architecture without VPN - Standard**

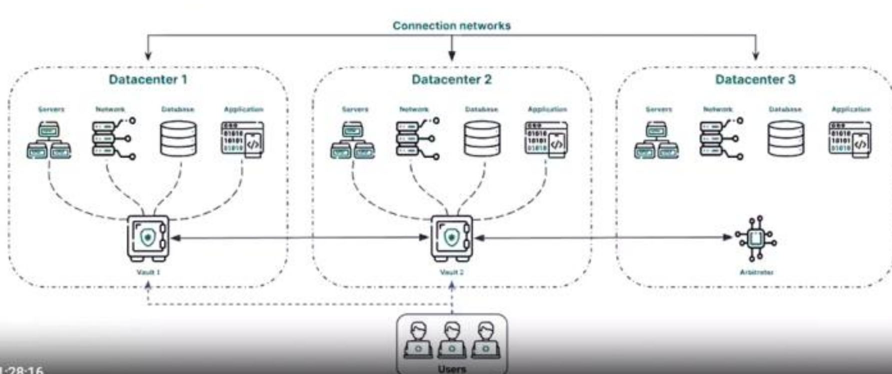
In this model, all connectivity is via the Network Connector. The main features of this architecture are:

- The senhasegura instance is not publicly accessible.
- The user can access senhasegura via the client's public IP.
- The client can allow more than one IP to access the senhasegura GUI.




# Architectures of senhasegura

**2 Instances of senhasegura + Arbitrator**



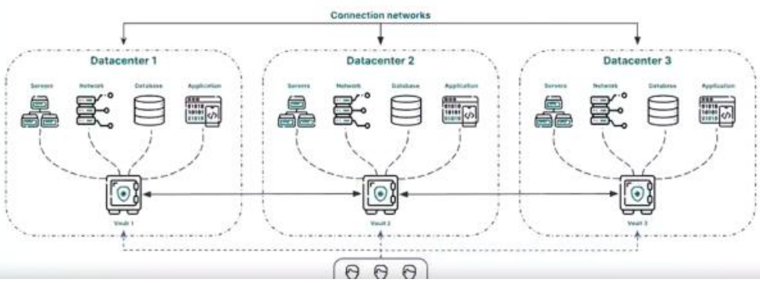
# Availability matrix

2 Instances of senhasegura + Arbitrator

| Scenario | Status | Automated Recovery | Additional Information |
|---|---|---|---|
| Instance 1 failure | Instance 2 | | If Instance 1 is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Failure of Instance 2 | Instance 1 | | If Instance 2 is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Failure of the link between datacenters | | | If the unavailable Instance is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Failure of Instance 1 and 2 | | | The master key process must be carried out. |

# Architectures of senhasegura

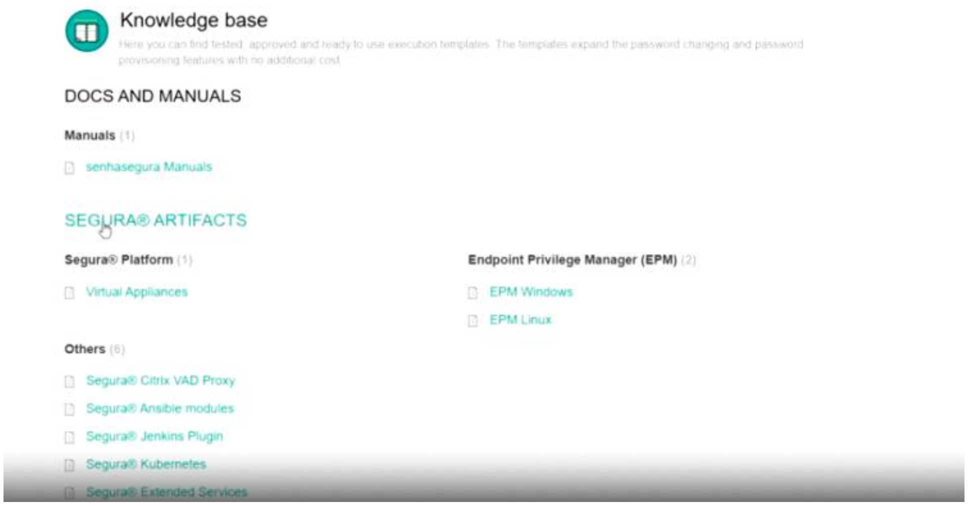
3 Instances of senhasegura

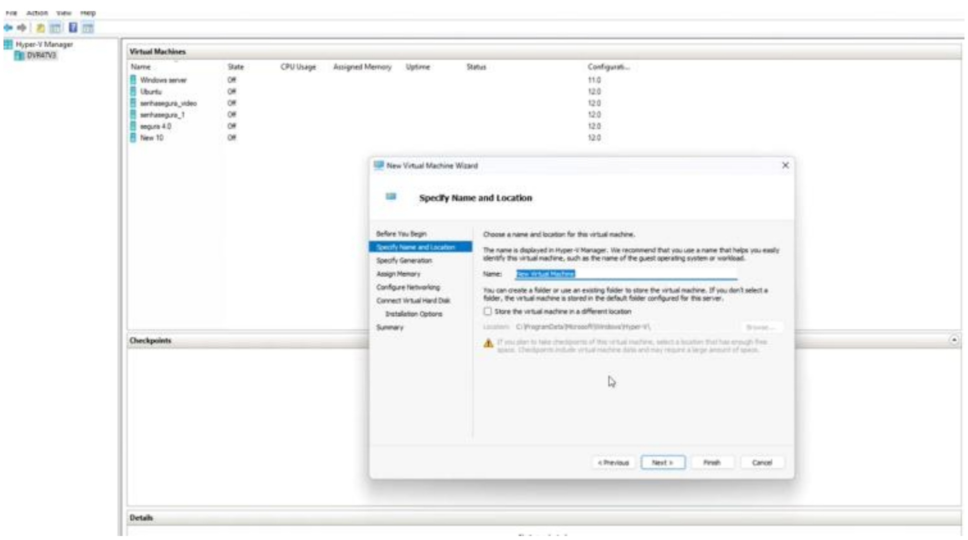


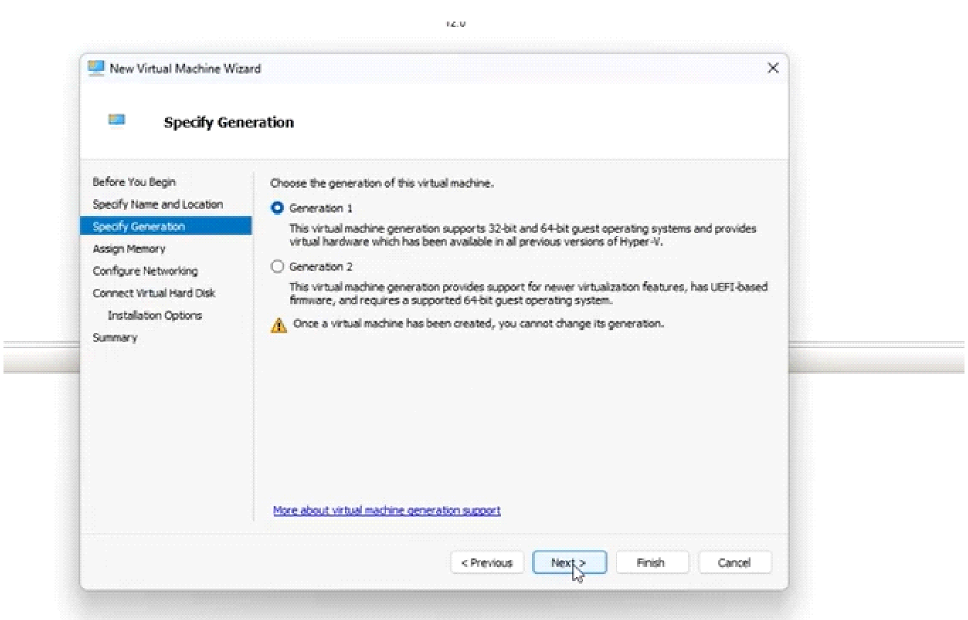

# Availability matrix

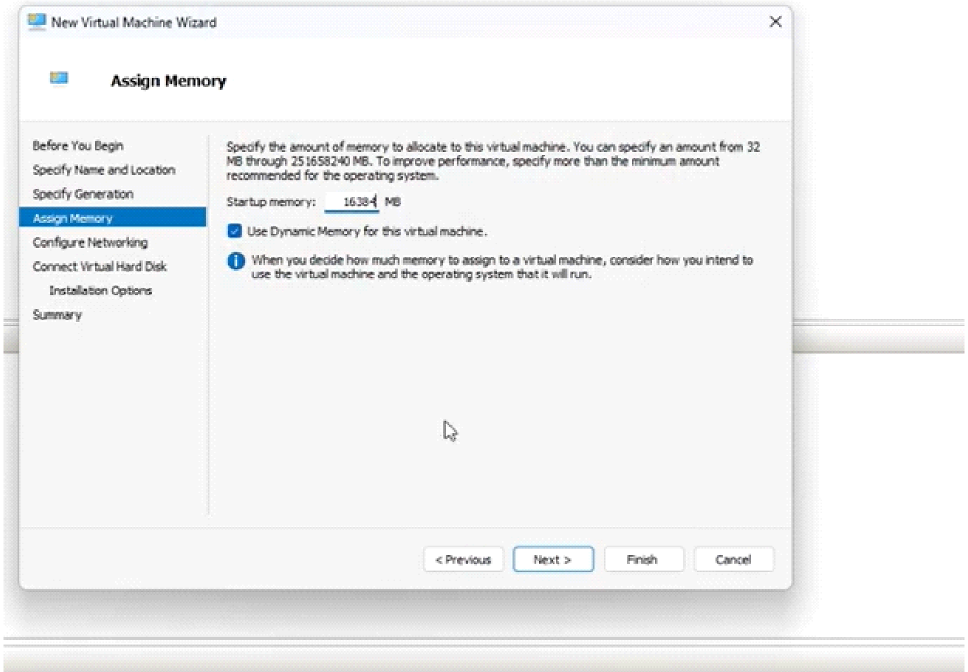
3 Instances of senhasegura

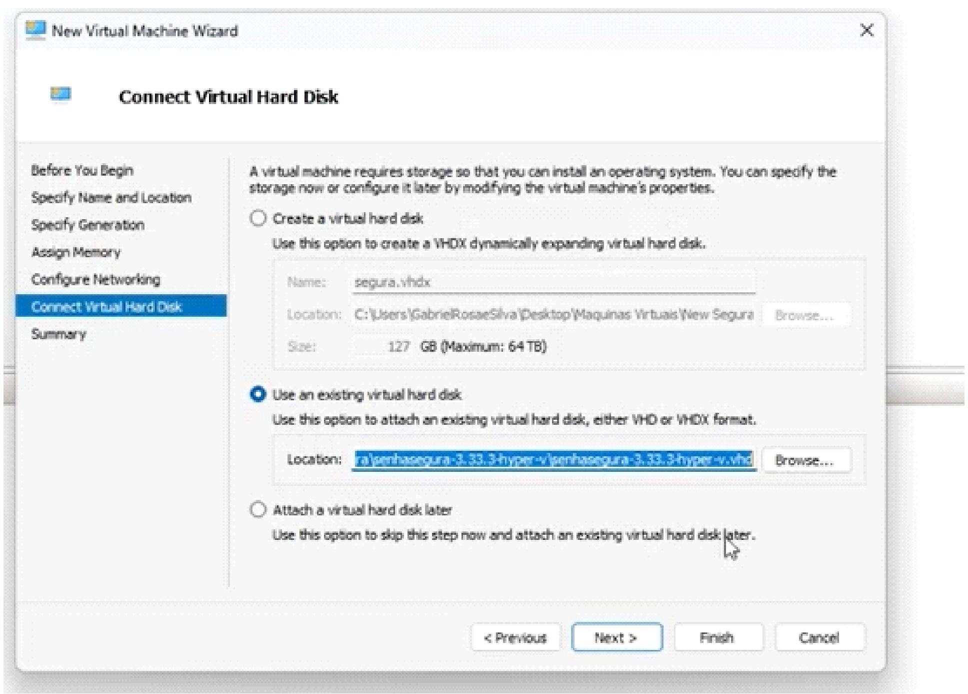
| Scenario | Status | Automated Recovery | Additional Information |
|---|---|---|---|
| Instance 1 failure | Instance 2 and 3 | | If Instance 1 is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Instance 2 failure | Instance 1 and 3 | | If Instance 2 is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Instance 3 failure | Instance 1 and 2 | | If Instance 3 is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Failure of link between datacenters | | | If the unavailable Instance is incommunicado for more than a week, resynchronization must be carried out by senhasegura Support. |
| Failure of Instance 1, 2 and 3 | | | The master key process must be carried out. |

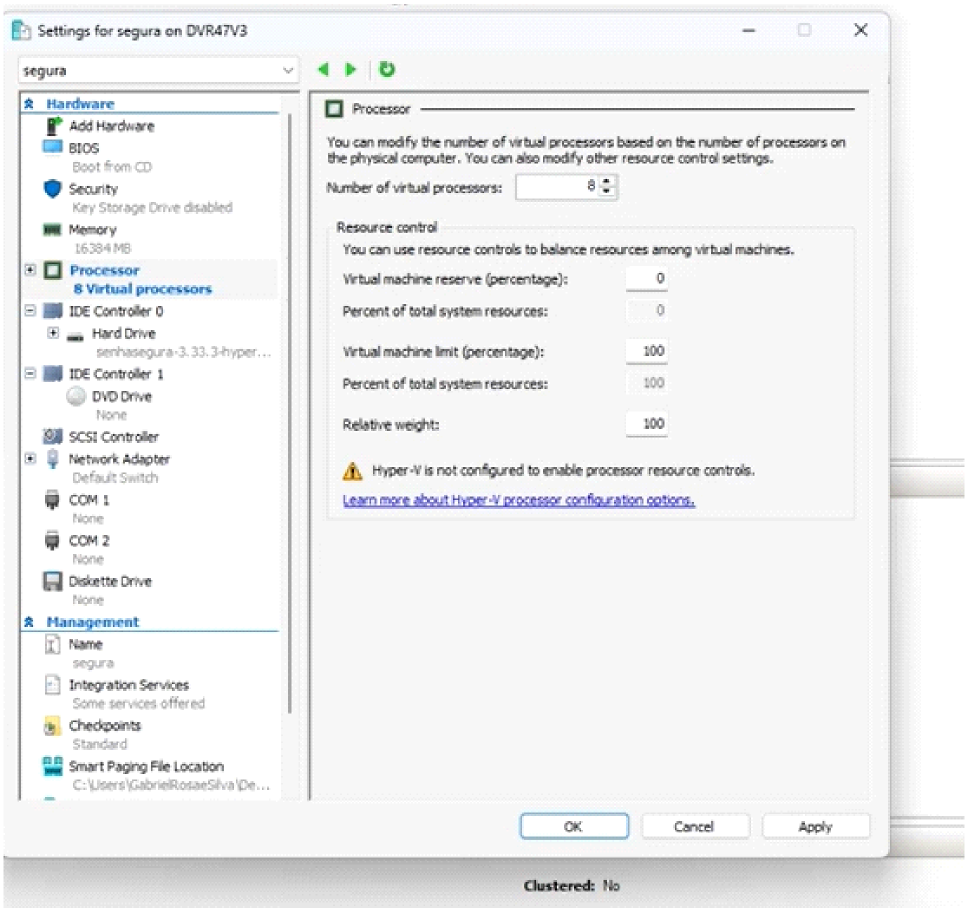

Knowledge base
DOCS AND MANUALS
Manuals (1)
senhasegura Manuals
SEGURA® ARTIFACTS
Segura® Platform (1)
Virtual Appliances
Endpoint Privilege Manager (EPM) (2)
EPM Windows
EPM Linux
Others (6)
Segura® Citrix VAD Proxy
Segura® Ansible modules
Segura® Jenkins Plugin
Segura® Kubernetes
Segura® Extended Services


Virtual Machines
New Virtual Machine Wizard
Specify Name and Location
Checkpoints
Details

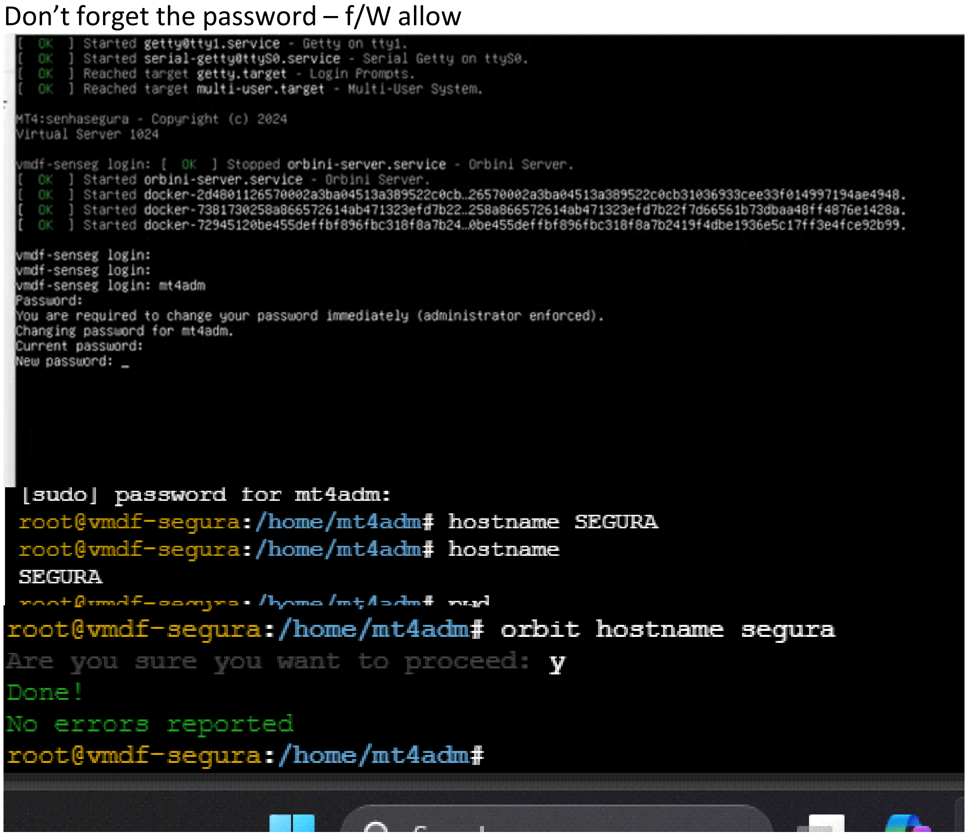

New Virtual Machine Wizard
Specify Generation
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Choose the generation of this virtual machine.
Generation 1
This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.
Generation 2
This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.
Once a virtual machine has been created, you cannot change its generation.
More about virtual machine generation support
< Previous
Next >
Finish
Cancel

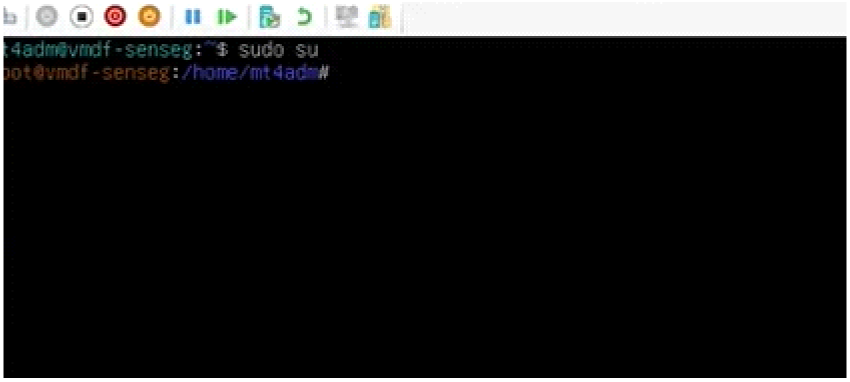

New Virtual Machine Wizard
Assign Memory
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary
Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 251658240 MB. To improve performance, specify more than the minimum amount recommended for the operating system.
Startup memory: 16384 MB
Use Dynamic Memory for this virtual machine.
When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.
< Previous
Next >
Finish
Cancel

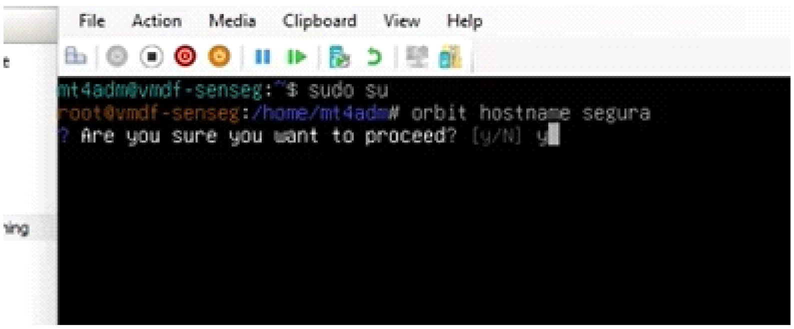

New Virtual Machine Wizard
Connect Virtual Hard Disk
Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Summary
A virtual machine requires storage so that you can install an operating system. You can specify the storage now or configure it later by modifying the virtual machine's properties.
Create a virtual hard disk
Use this option to create a VHDX dynamically expanding virtual hard disk.
Name: segura.vhdx
Location: C:\Users\GabrielRosaeSilva\Desktop\Maquinas Virtuais\New Segura
Browse...
Size: 127 GB (Maximum: 64 TB)
Use an existing virtual hard disk
Use this option to attach an existing virtual hard disk, either VHD or VHDX format.
Location: ra\senhasegura-3.33.3-hyper-v\senhasegura-3.33.3-hyper-v.vhd
Browse...
Attach a virtual hard disk later
Use this option to skip this step now and attach an existing virtual hard disk later.
< Previous
Next >
Finish
Cancel



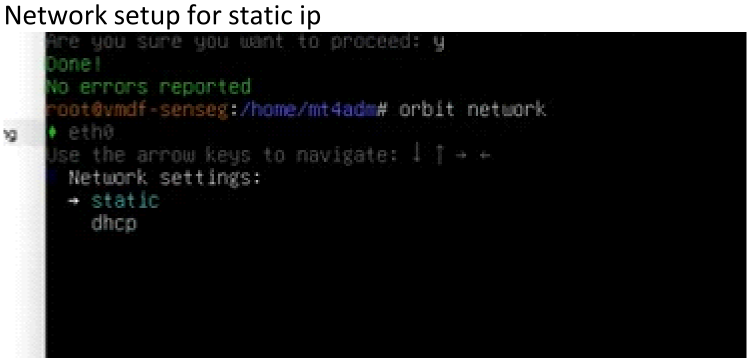

Don't forget the password – f/W allow





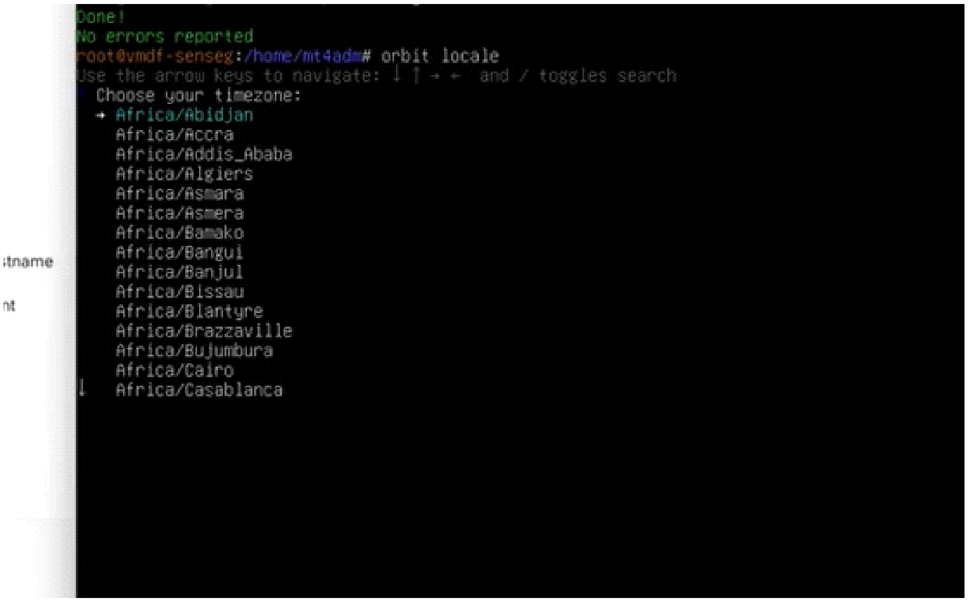



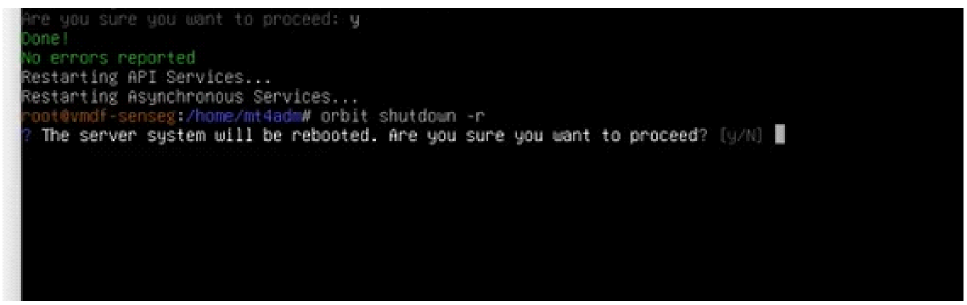

Network setup for static ip



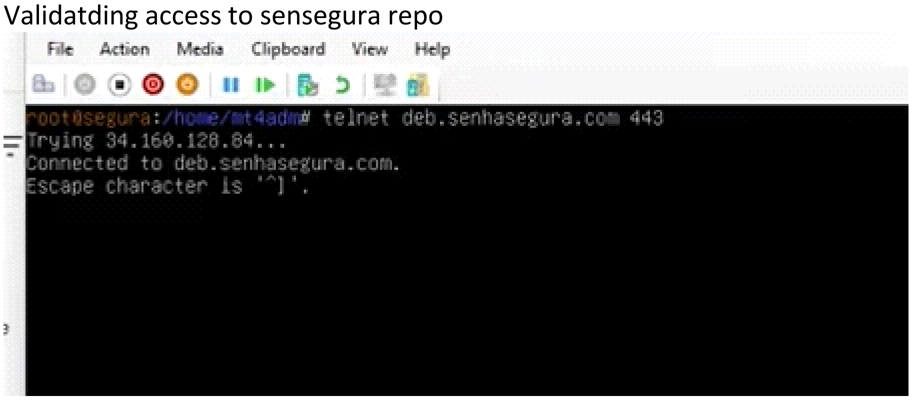


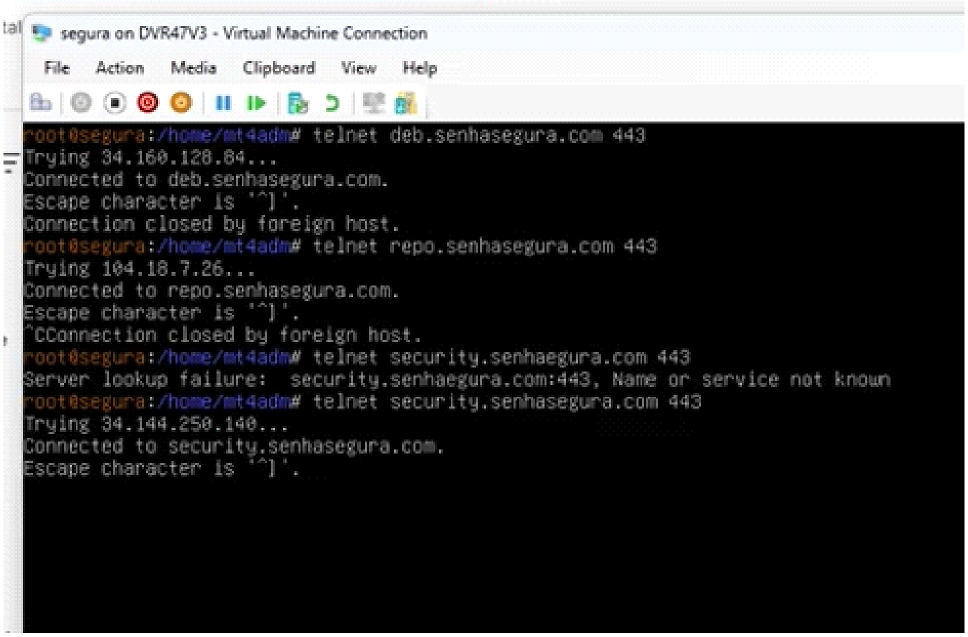
timzone




Validatding access to sensegura repo

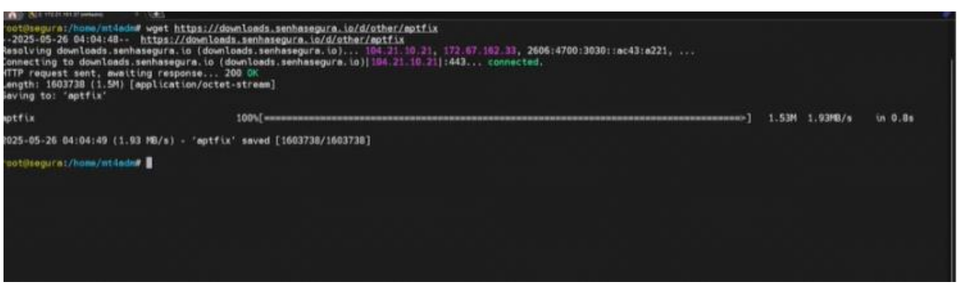




Before shutdown make sure to take a snapshot



Updating Segura instance




```
2025-05-26 04:04:49 (1.93 MB/s) - 'aptfix' saved [1603738/1603738]

root@segura:/home/mt4adm# sudo chmod +x aptfix
root@segura:/home/mt4adm# sudo ./aptfix

Duration: 150.017371ms
 (Reading database ... 241843 files and directories currently installed.)
 Preparing to unpack /tmp/apt-fix.deb ...
 Unpacking apt-fix (1.0.0-3) over (1.0.0-3) ...
 Setting up apt-fix (1.0.0-3) ...
 Adding mt4 repository key... Cleaning apt cache...
 Registering ca certificate...
 Updating certificates in /etc/ssl/certs...
```

```
Duration: 5.327216508s

Duration: 1.791945ms
root@segura:/home/mt4adm# apt update
Get:1 https://deb.senhasegura.com/stable-sp bookworm InRelease [15.0 kB]
Get:2 https://deb.senhasegura.com/stable-sp bookworm-updates InRelease [12.5 kB]
Get:3 https://deb.senhasegura.com/stable-sp bookworm-security InRelease [12.6 kB]
Get:4 https://deb.senhasegura.com/stable-sp bookworm/main amd64 Packages [456 kB]
Get:5 https://deb.senhasegura.com/stable-sp bookworm/extras amd64 Packages [126 kB]
Get:6 https://deb.senhasegura.com/stable-sp bookworm/senhasegura amd64 Packages [68.2 kB]
Get:7 https://deb.senhasegura.com/stable-sp bookworm-updates/main amd64 Packages [2,151 B]
Get:8 https://deb.senhasegura.com/stable-sp bookworm-security/main amd64 Packages [32.9 kB]
Fetched 725 kB in 3s (262 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

0:59 / 2:44

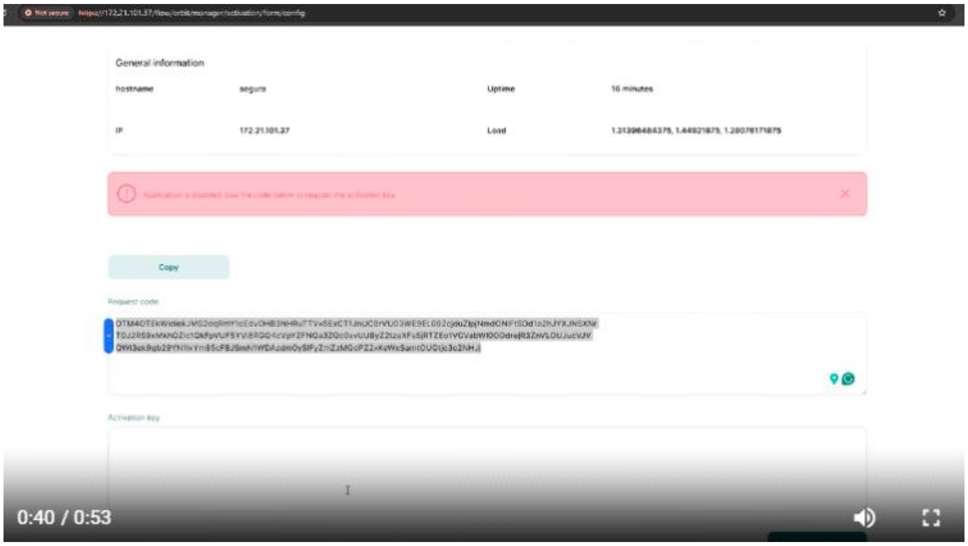
```
orbit-cli is already the newest version (4.0.7-12~bookworm).
0 upgraded, 0 newly installed, 0 to remove and 220 not upgraded
root@segura:/home/mt4adm#
root@segura:/home/mt4adm# orbit update
IMPORTANT

The recommended minimum requirement for a reasonable system wor
```

```
root@segura:/home/mt4adm# orbit version
senhasegura Orbit Command-Line Interface
Version 4.0.7-12

+------------------------+---------+
|  APPLICATION PACKAGES  | VERSION |
+------------------------+---------+
| orbini-framework       | 4.0.0-2 |
| senhasegura            | 4.0.0-2 |
| senhasegura-backend    | 4.0.0-2 |
| senhasegura-proxy      | 4.0.0-2 |
| senhasegura-proxy-utils| 4.0.0-2 |
| senhasegura-proxy-v1   | 4.0.0-2 |
| senhasegura-utils      | 4.0.0-2 |
+------------------------+---------+
root@segura:/home/mt4adm#
```

```
5b9a7f94e1acdfa720bb3f28 to empty string
 Loaded image: selenium-senhasegura-image:latest
 Setting up senhasegura-proxy-db-postgresql (4.0.18-1~bookworm) ...
 Loading docker image.
9b8b107d97e2: Loading layer  838.3MB/838.3MB
Loaded image: senhasegura-proxy-db-postgresql:4.0.18-1
 Installing systemctl service.
 Checking proxy-db-postgresql SSL configuration...
 Created symlink /etc/systemd/system/default.target.wants/proxy-db-postgresql.service → /lib/s
 Setting up taco-senhasegura (4.0.20-1~bookworm) ...
 Setting up orbini-coda (4.0.16-1~bookworm) ...
 [2026-03-04 14:48:29]: Initiating the installation orbini-coda
 [2026-03-04 14:48:42]: Installed successfully!
 Setting up cosh-senhasegura-image (4.0.17-1~bookworm) ...
 Loading docker image.
```

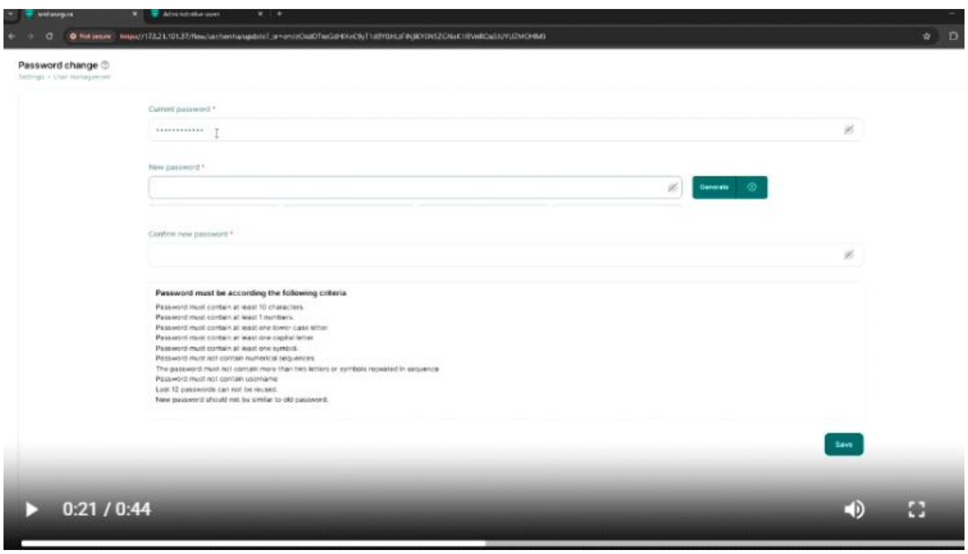
# activation

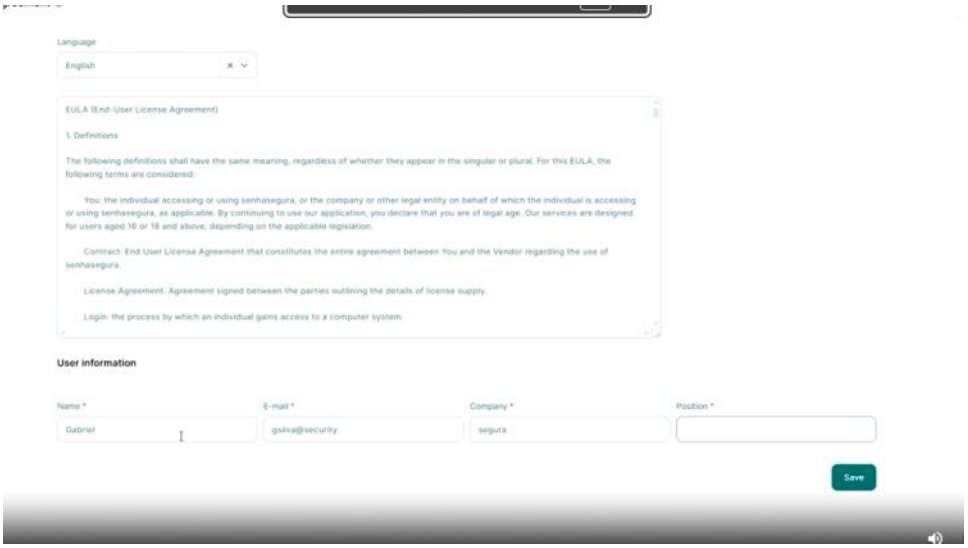
Wednesday, March 04, 2026 2:45 PM

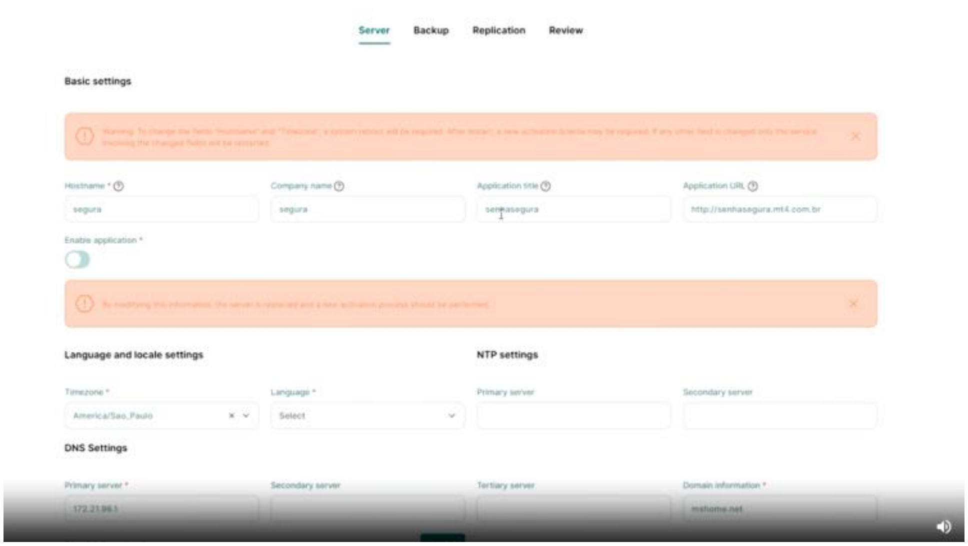


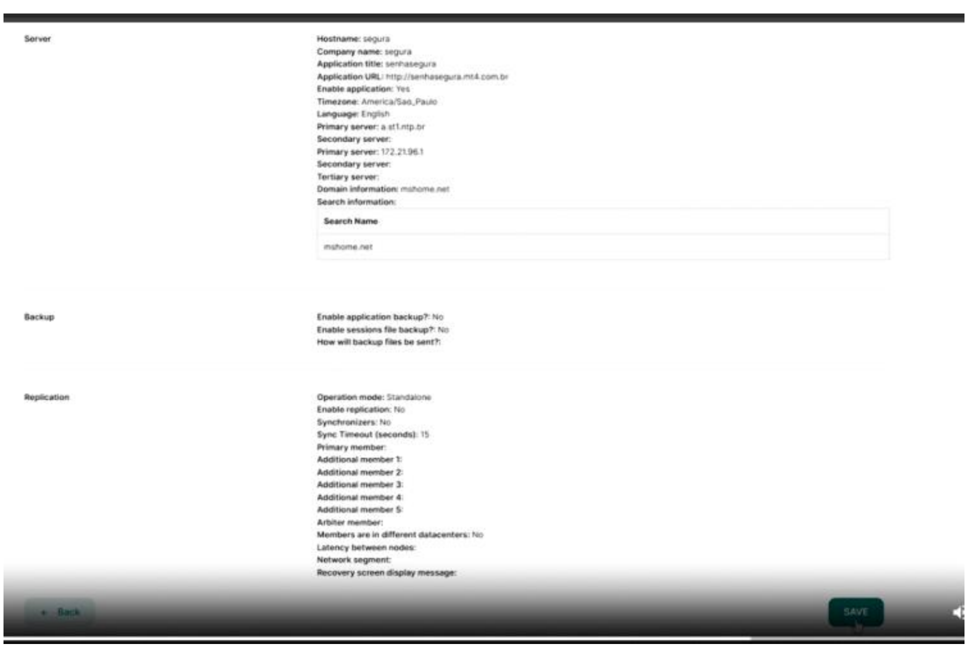

[:: Affinity ::]()

mt4web






Default passwords :About Administrative users he core PAM solution (Privileged Access Management) is built out of multiple components that work together. Typically, these are all included in the virtual appliance image you deployed. For example, you have:

1. The **PAM Core** – This is the main engine that handles privileged access management workflows, policies, and approvals.
2. **Orbit** – That's the control console you're using to manage the system (like starting services, checking status, etc.).
3. **Proxy** – This handles secure connections between clients and the backend.
4. **Backend** – This is the database and business logic layer storing user credentials, sessions, etc.

successfully deployed the senhasegura appliance in Azure, and both the backend services and web interface are up and running. If you need further configuration—like setting up users, licenses, or resolving incidents—now you can proceed from that panel. Let me know if you need any help with the next steps!

"Where are these other components come? They all have lots of components, the PAM core and other things. So where are these other components come?"
00:08

Ah, great question! So, in senhasegura, the core PAM solution (Privileged Access Management) is built out of multiple components that work together. Typically, these are all included in the virtual appliance image you deployed. For example, you have:

5. The **PAM Core** – This is the main engine that handles privileged access management workflows, policies, and approvals.
6. **Orbit** – That's the control console you're using to manage the system (like starting services, checking status, etc.).
7. **Proxy** – This handles secure connections between clients and the backend.
8. **Backend** – This is the database and business logic layer storing user credentials, sessions, etc.

you've successfully deployed the senhasegura appliance in Azure, and both the backend services and web interface are up and running. If you need further configuration—like setting up users, licenses, or resolving incidents—now you can proceed from that panel. Let me know if you need any help with the next steps!

# Orbit web interface setup

Thursday, March 05, 2026 11:38 AM

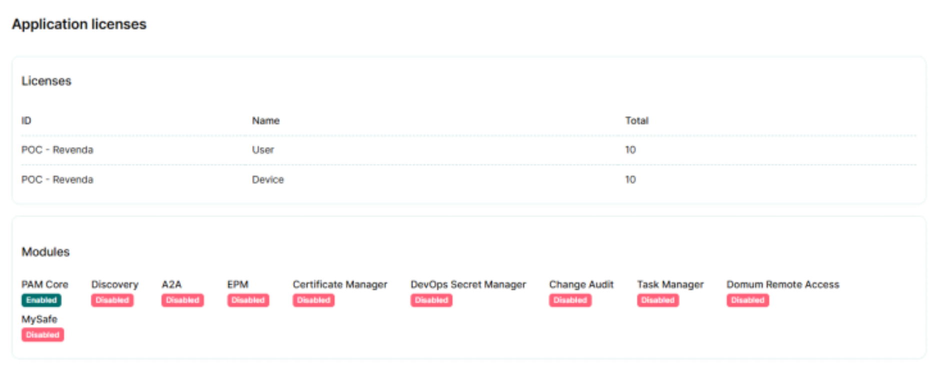
## General Information

This section provides general information about the server. It includes the hostname, IP address, server uptime, and the number of available CPUs on that server. You can also find the server's average workload.

# PAM core- devise onboarding & credential mapping

Thursday, March 05, 2026 11:14 AM

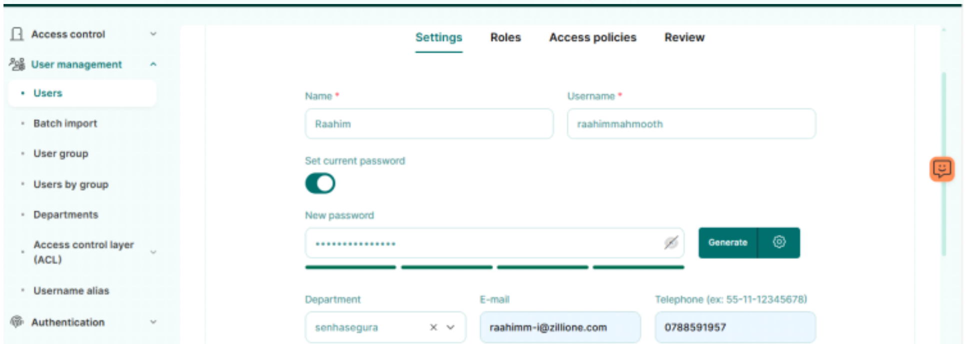
1. Going with **mt4adm** will not be secure unless there is a emergency so create a user and assigned roles and assigne rights that perticular role

- Central console – settings – users
- User have history, can be forgotten in that case user name and password will filled with random values
-
- 

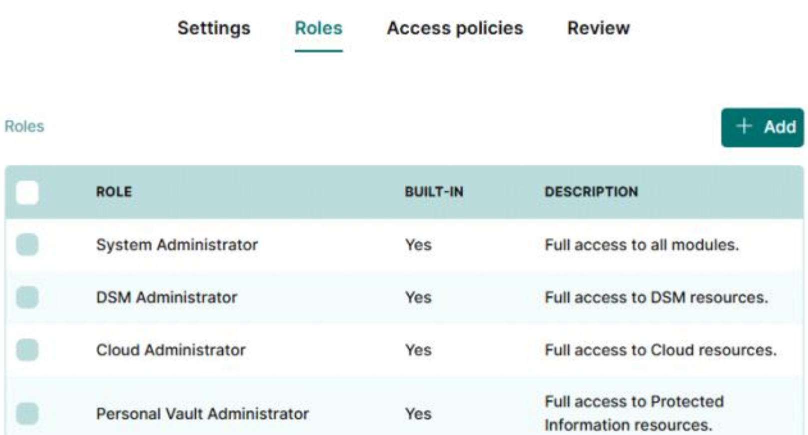

**After deploying senhasegura, some applications (like Home, Settings, etc.) stay disabled until the Orbit services are initialized.**

```
orbit: error: unexpected argument status
root@segura:/home/mt4adm# orbit app status

Application: Inactive
Replication: Inactive
Instance:    Primary
Node Name:   segura_2886729732
Maintenance: No
Main:        Yes

root@segura:/home/mt4adm#
```

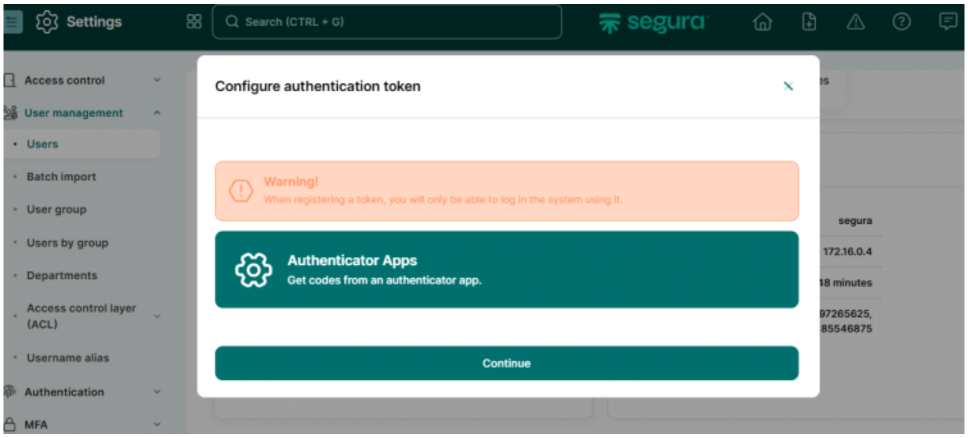
```
root@segura:/home/mt4adm# orbit app start
The application services will be stopped or restarted during the process. Are you sure you want to proceed: y

Application: Active
Replication: Inactive
Instance:    Primary
Node Name:   segura_2886729732
Maintenance: No
Main:        Yes
```

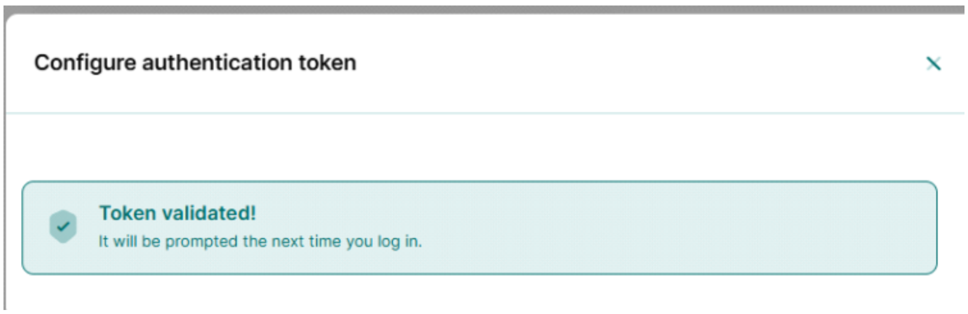




Roles




Configure mfa

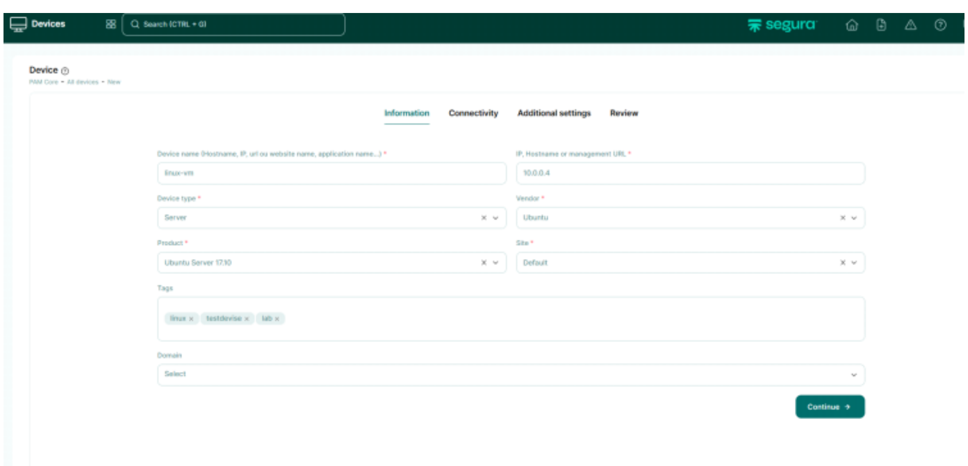


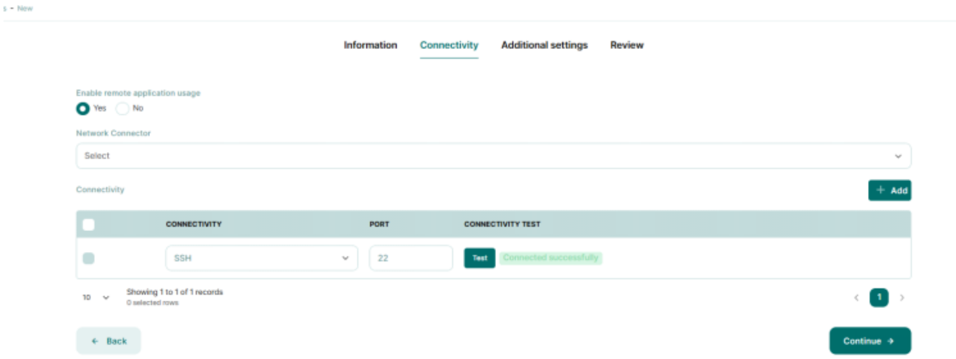

## What you are doing Devise onboarding

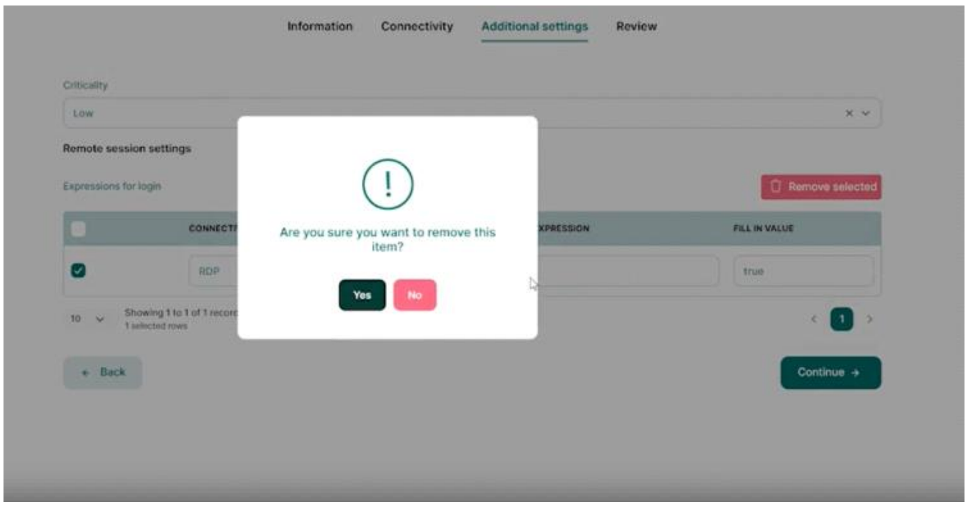
You are adding a **target machine (asset)** into PAM so you can later:

- store credentials
- connect (RDP/SSH)
- manage privileged access

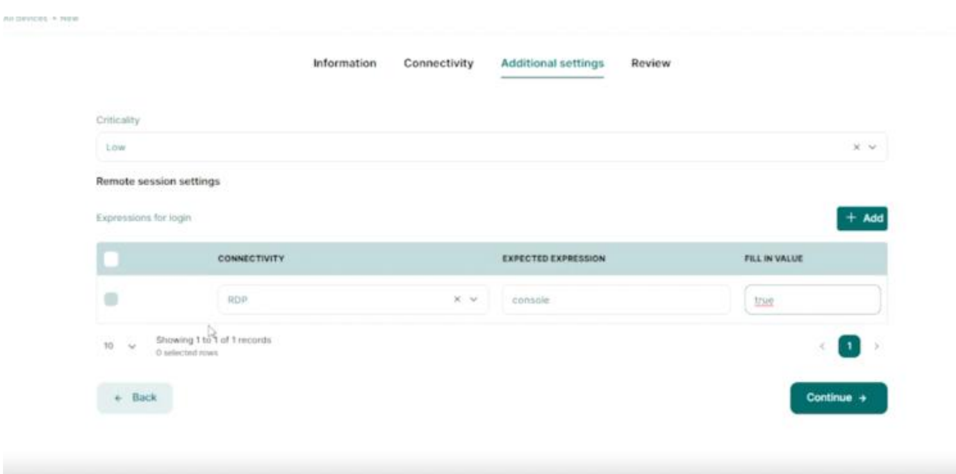



CONNECTIVITY

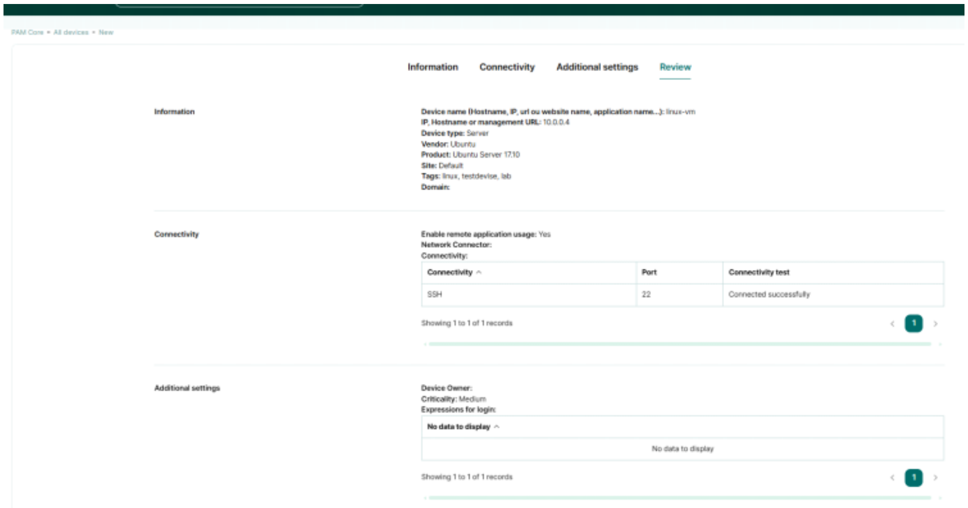


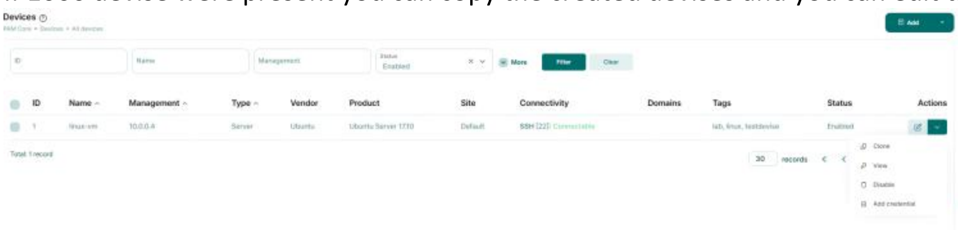

## Additional settings



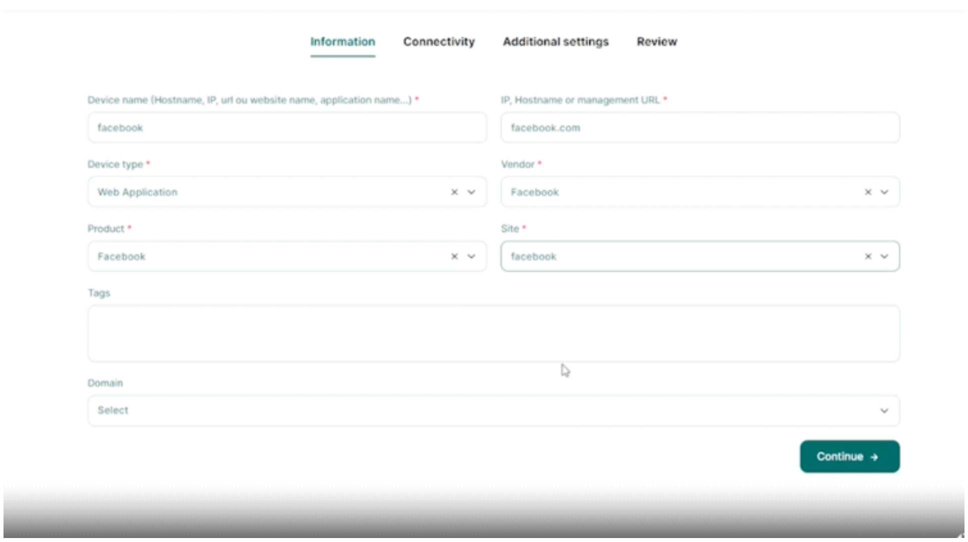

## Review

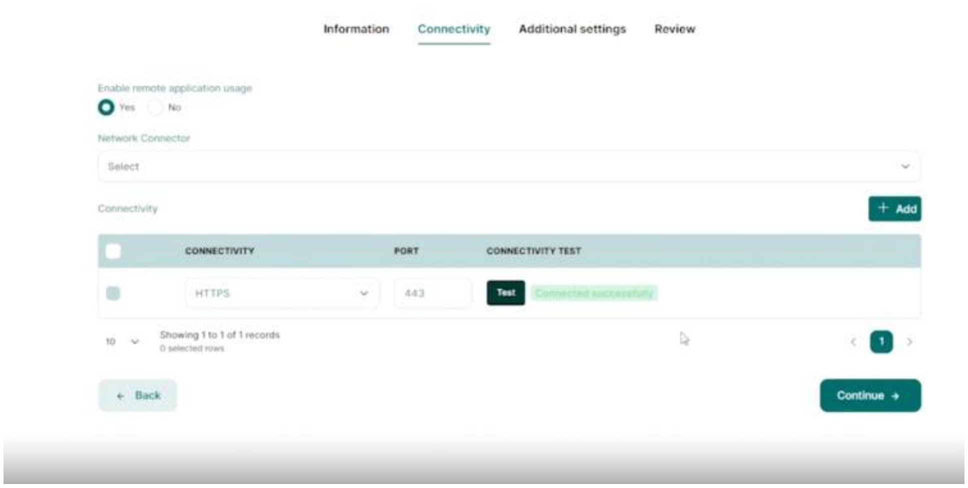



If 1000 devise were present you can copy the created devises and you can edit the configuration like Ip




If to access a web application



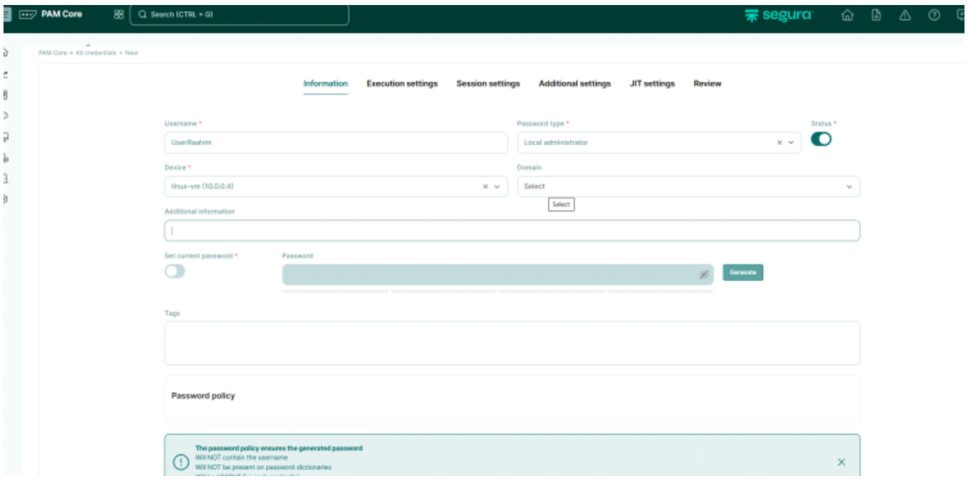


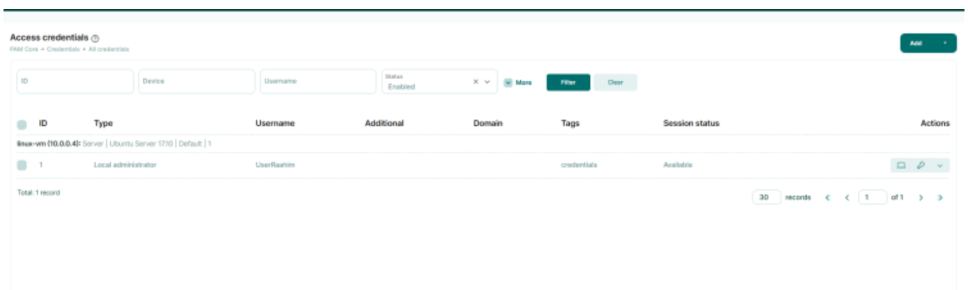

Credential Creation

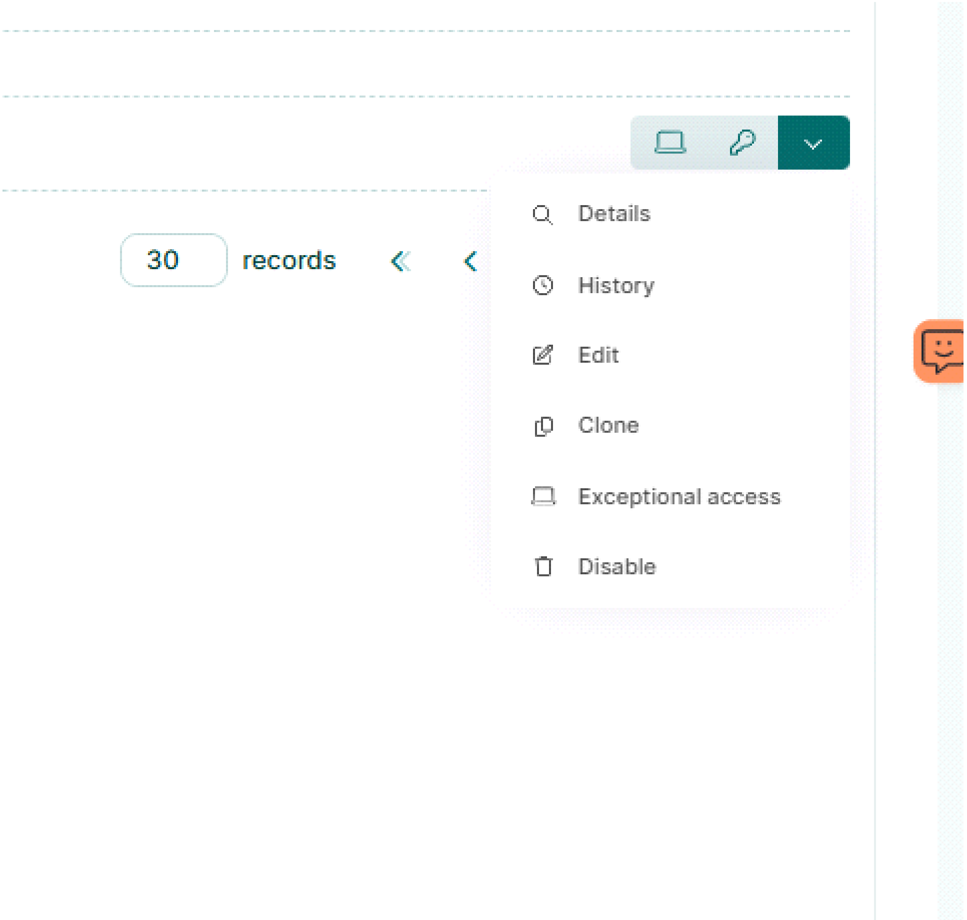
in a real-world setup, manually collecting and configuring every user's password isn't practical. Instead, organizations usually rely on centralized identity management. For example, tools like LDAP, Active Directory, or even SSO (Single Sign-On) systems. In that setup, users authenticate once, and PAM checks against that central directory rather than individual passwords. So, once users onboard their devices and get linked to the central system, PAM just references that directory. This way, users only manage one set of credentials, and the admin side scales much better.

essentially mapping the device (or user) to a central set of credentials stored in a directory or identity provider. Once that mapping is done, PAM just uses that central identity to validate access—so it scales, it stays consistent, and users only need to remember one set of credentials across all their devices.

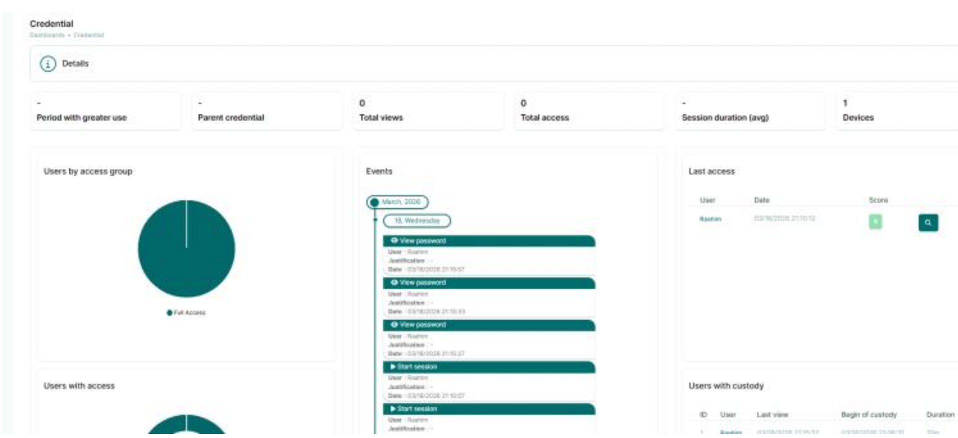
Note : since we don't have a central identity manager we set the current password or will make that to use the existing password of the same Linux machine





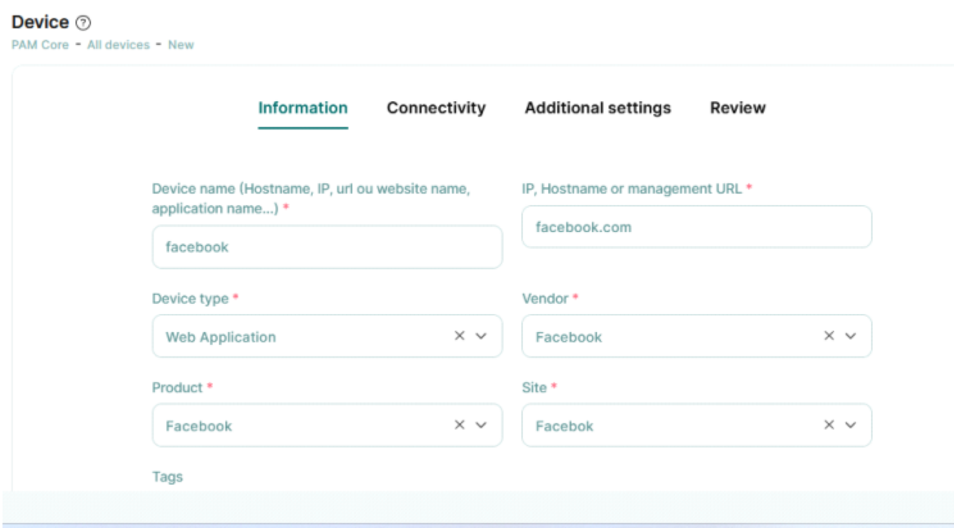

History and events who vied who stated sessions



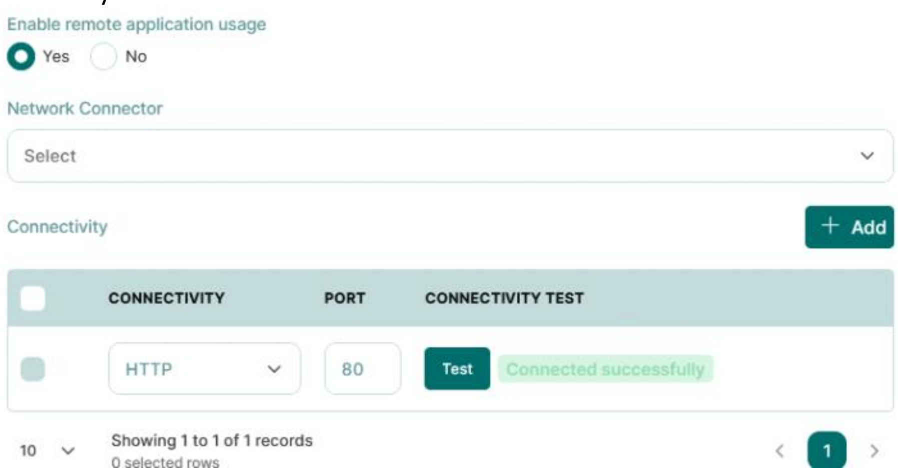
# Web session automation

Monday, March 30, 2026 10:20 AM

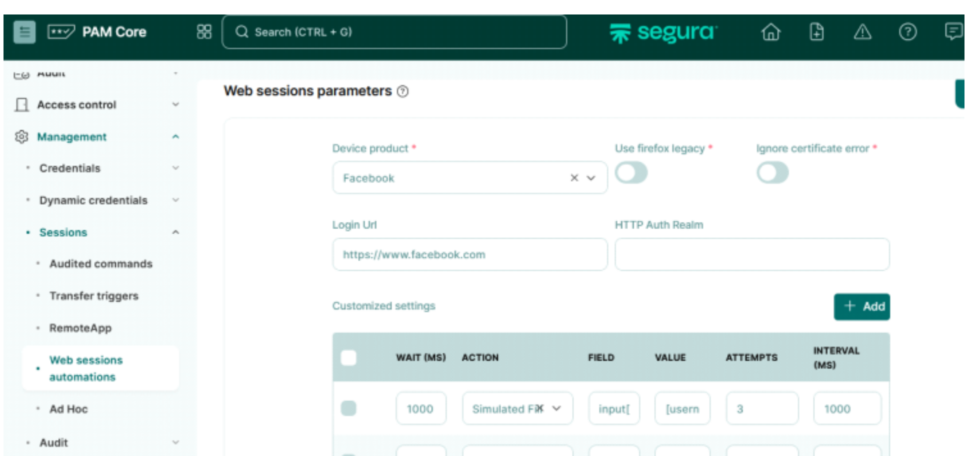
## Create a Devise




## Connectivity




## Web session automation

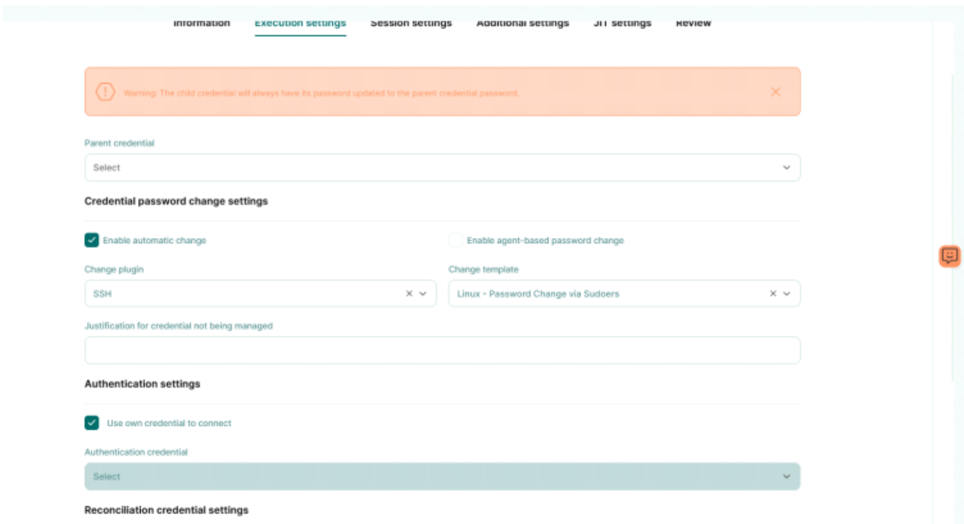


# Password rotation

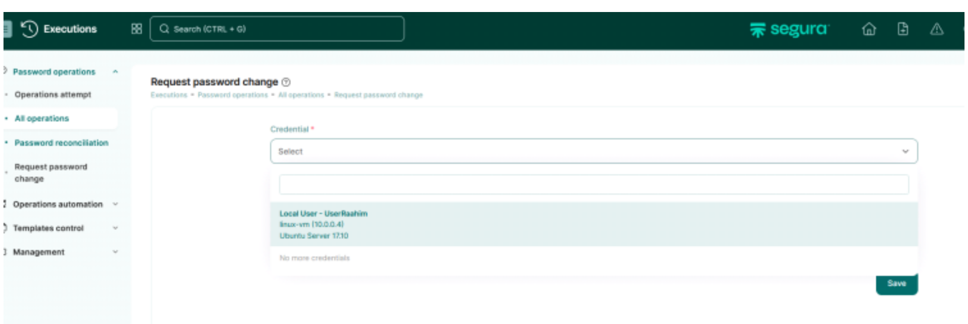
Monday, March 30, 2026     3:16 PM

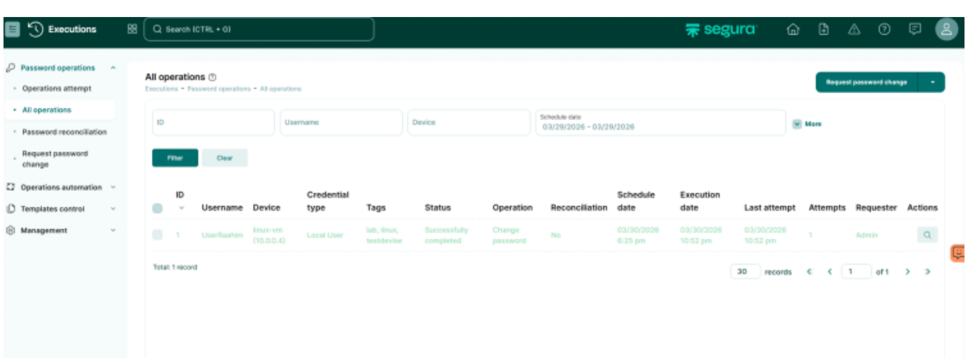
## Linux

- Password change request – the automation to setup on a Linux machine – credential mapping – change template and plugin setup

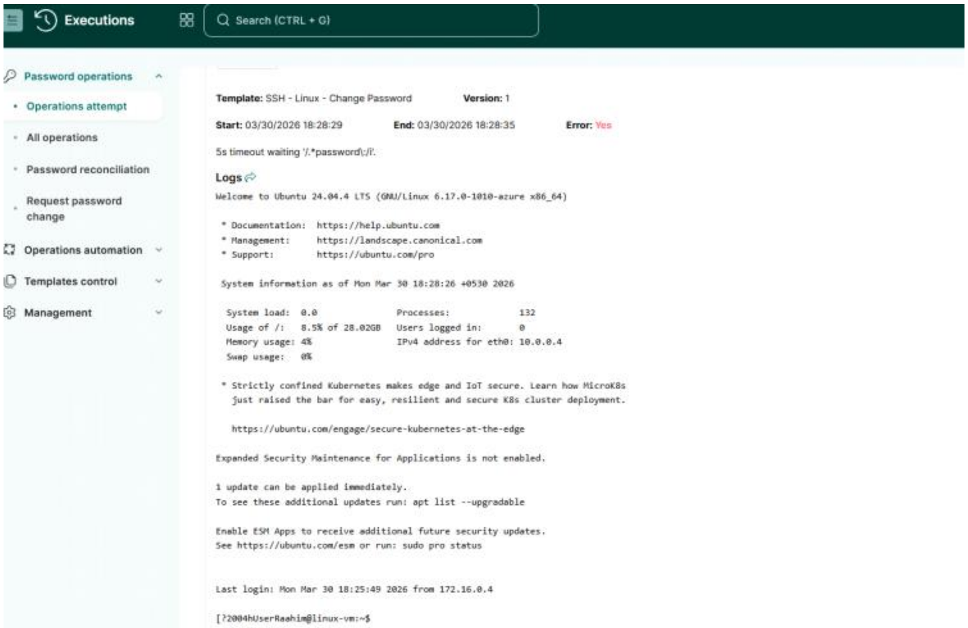



- Request password change



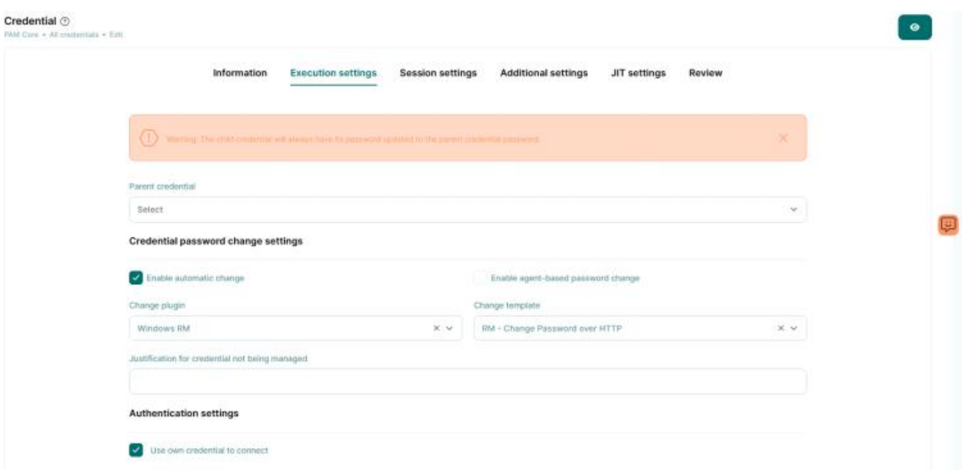




- If any error occurs

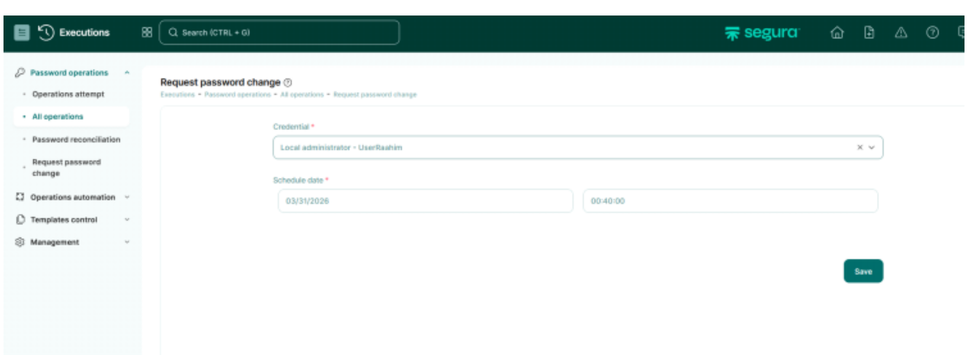



# Windows

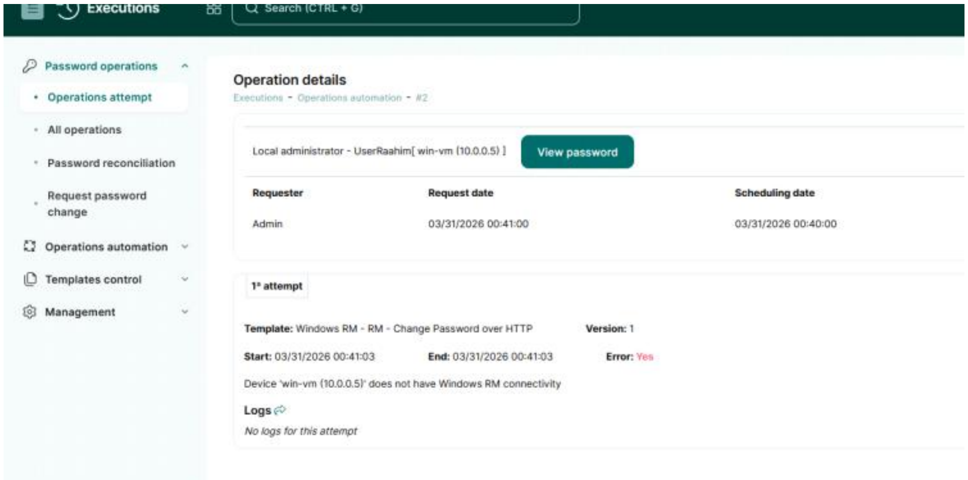
- Execution settings – windows uses different templates in order for remote excecution



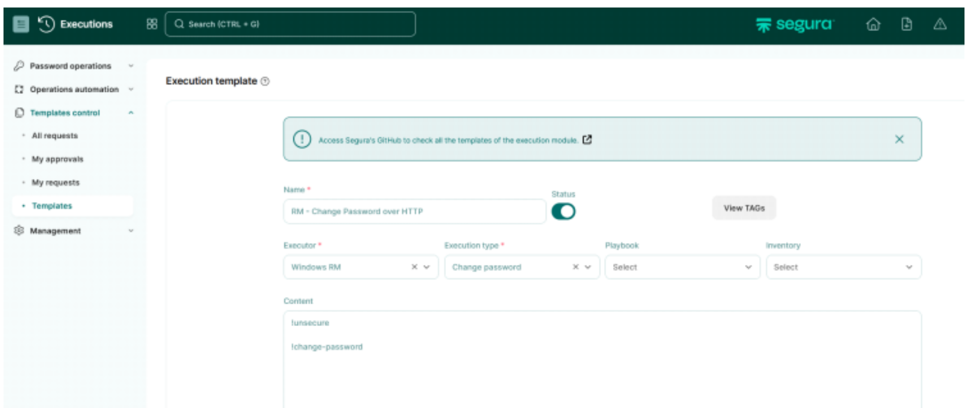

- Request password change



Error behavior

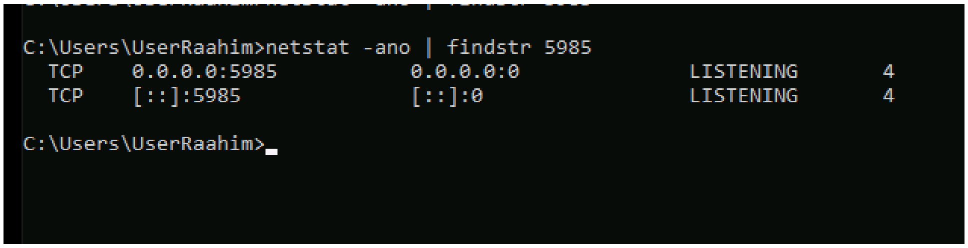


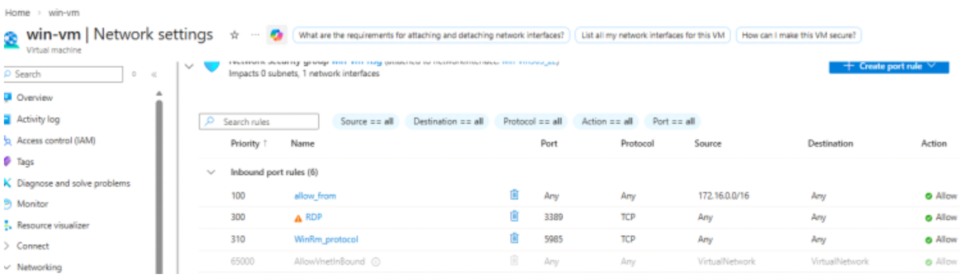

Password rotation template




WinRm config -- see the full commands

```
PS C:\Users\UserRaahim> winrm quickconfig
WinRM is not set up to receive requests on this machine.
The following changes must be made:

Start the WinRM service.
Set the WinRM service type to delayed auto start.

Make these changes [y/n]? y

WinRM has been updated to receive requests.

WinRM service type changed successfully.
WinRM service started.
WSManFault
    Message
        ProviderFault
            WSManFault
                Message = WinRM firewall exception will not work since one of the network connection types on this m
ne is set to Public. Change the network connection type to either Domain or Private and try again.

Error number:  -2144108183 0x80338169
WinRM firewall exception will not work since one of the network connection types on this machine is set to Public. C
e the network connection type to either Domain or Private and try again.
PS C:\Users\UserRaahim>
```

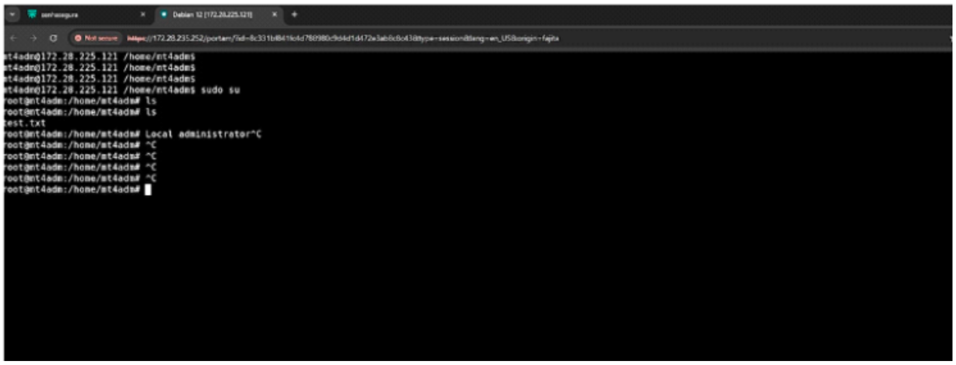

C:\Users\UserRaahim>netstat -ano | findstr 5985
TCP 0.0.0.0:5985 0.0.0.0:0 LISTENING 4
TCP [::]:5985 [::]:0 LISTENING 4
C:\Users\UserRaahim>


Home > win-vm
win-vm | Network settings
Virtual machine
What are the requirements for attaching and detaching network interfaces?
List all my network interfaces for this VM
How can I make this VM secure?
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Impacts 0 subnets, 1 network interfaces
Create port rule
Search rules
Source == all
Destination == all
Protocol == all
Action == all
Port == all
Priority
Name
Port
Protocol
Source
Destination
Action
Inbound port rules (6)
100 allow_from Any Any 172.16.0.0/16 Any Allow
300 RDP 3389 TCP Any Any Allow
310 WinRm_protocol 5985 TCP Any Any Allow
65000 AllowVnetInBound Any Any VirtualNetwork VirtualNetwork Allow

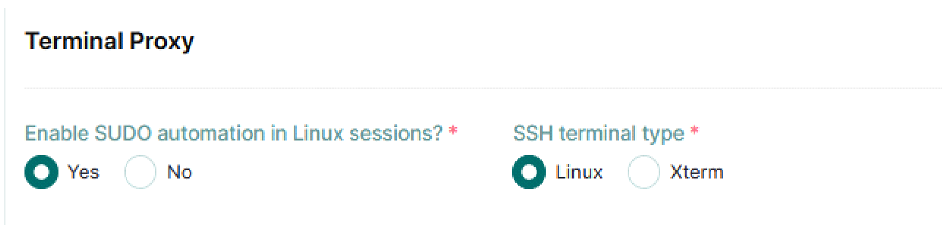
# Session establishment methods

Monday, March 30, 2026 3:09 PM

Running a session

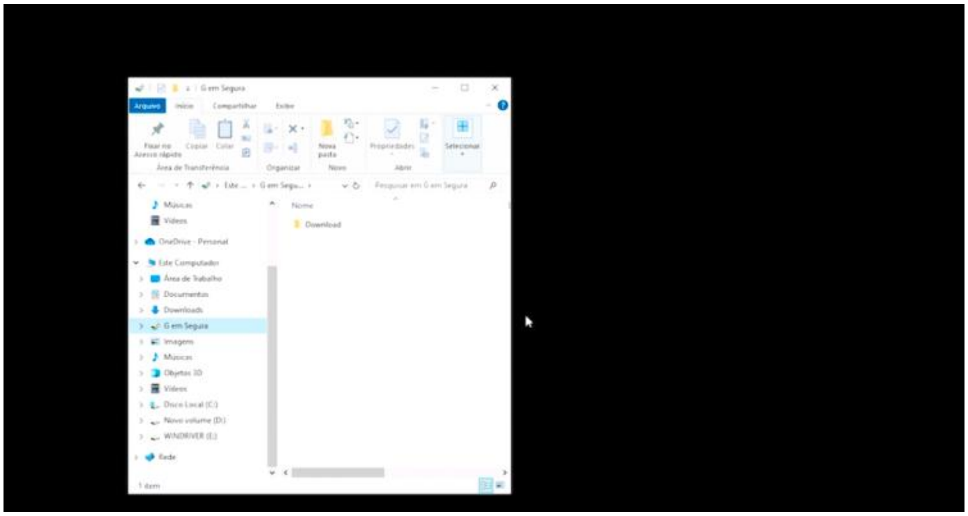



Enable sudo automation in ssh session in settings-->global parameters

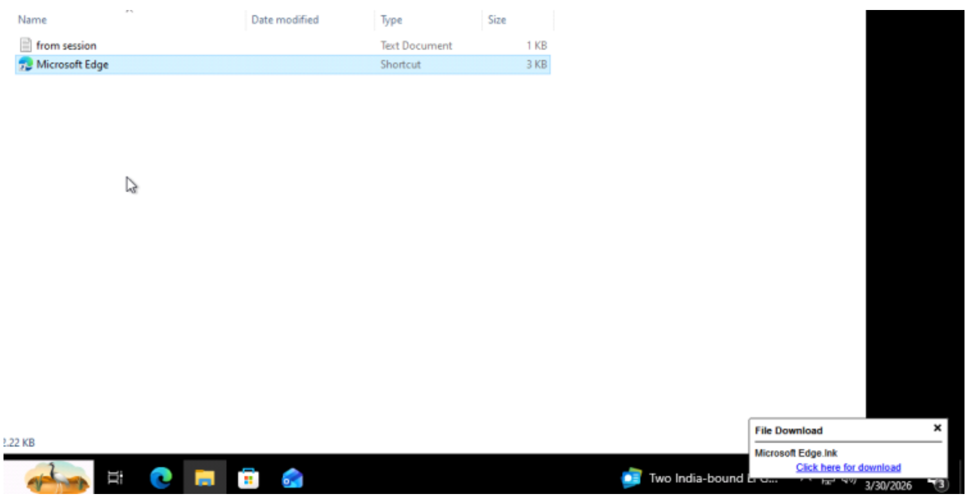



But before stating web session you need to configure the web automation template


**Rdp**



- You can simply drag and drop the file
- You can enable cntl+v or +c can enable
- To download you need to move it in download folder it will create a link
- Normally ssh or web session can be terminated by closing the tab but in the rdp you need to sign out or disconnect
-
-

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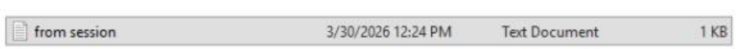


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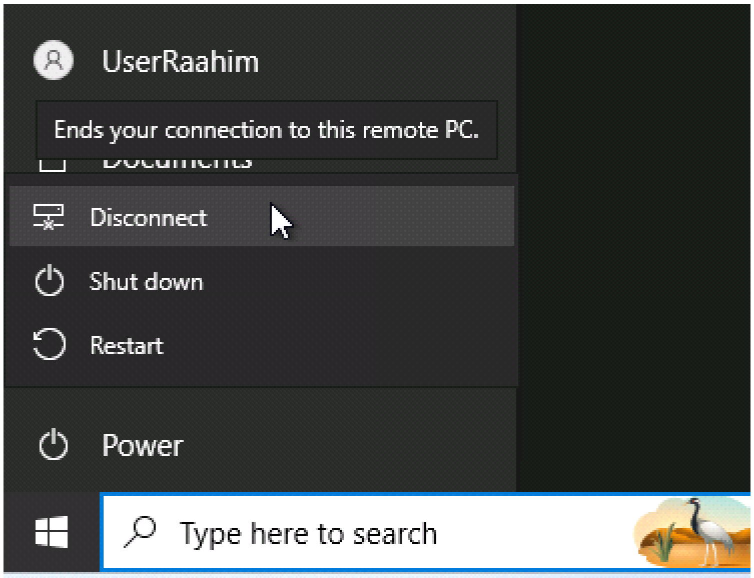
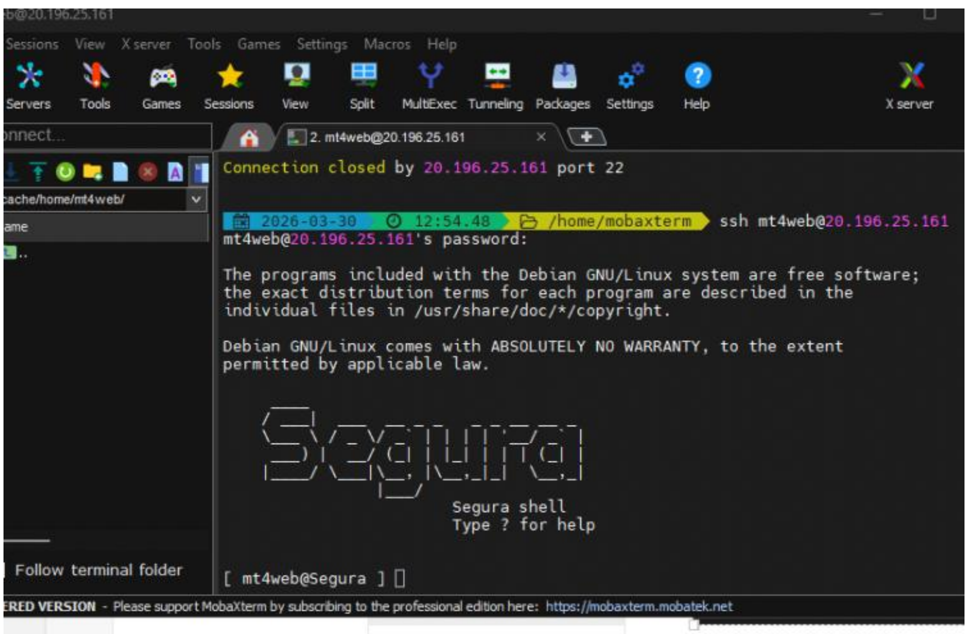
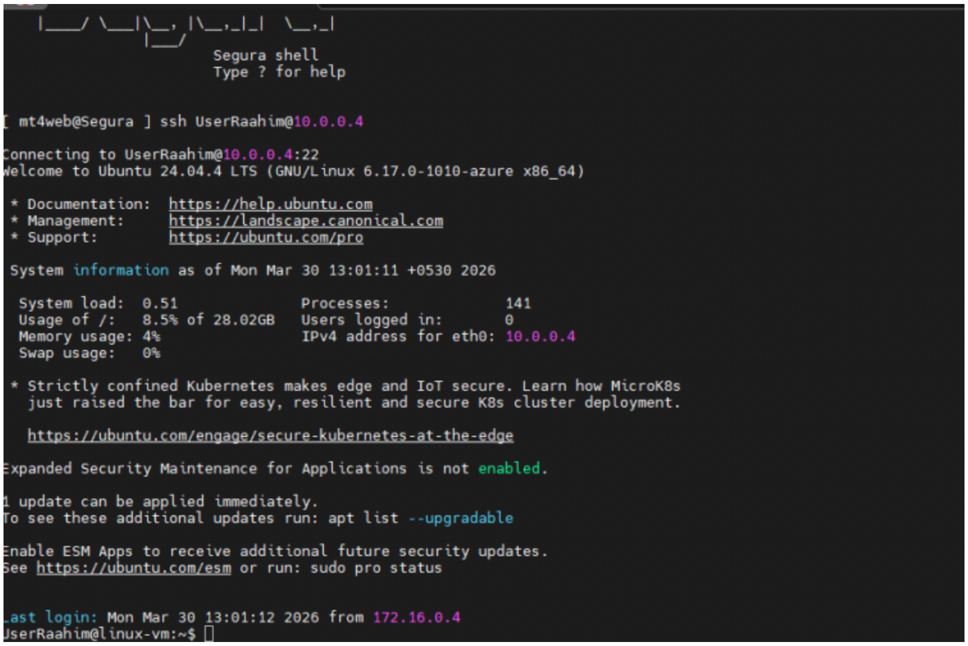


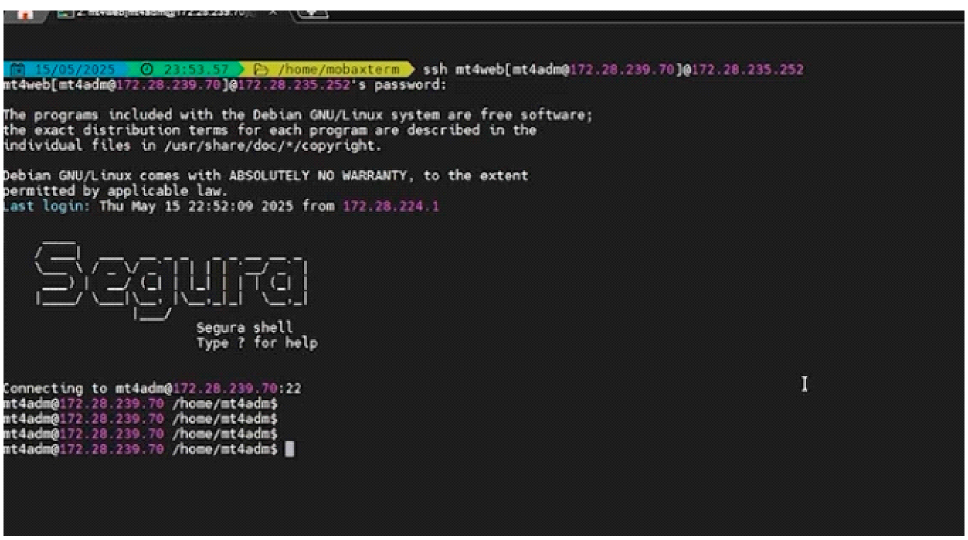

**You can initialize a session not only through web instead mobaXterm means getting access to proxy terminal -- ssh**



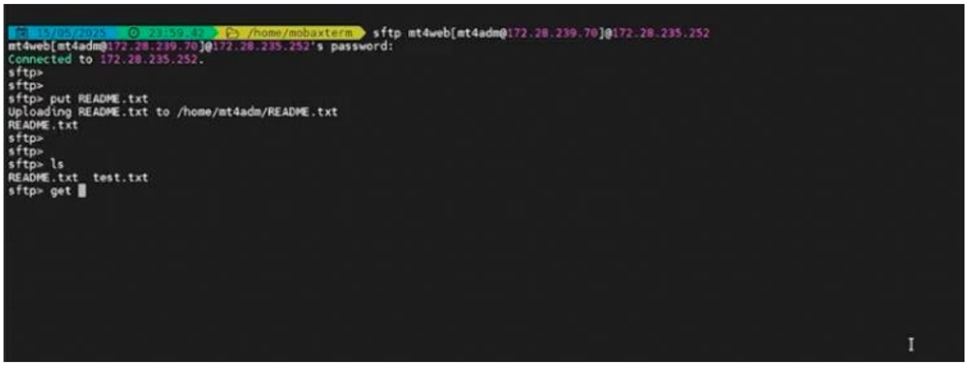



You can establish a session from there to devise by this

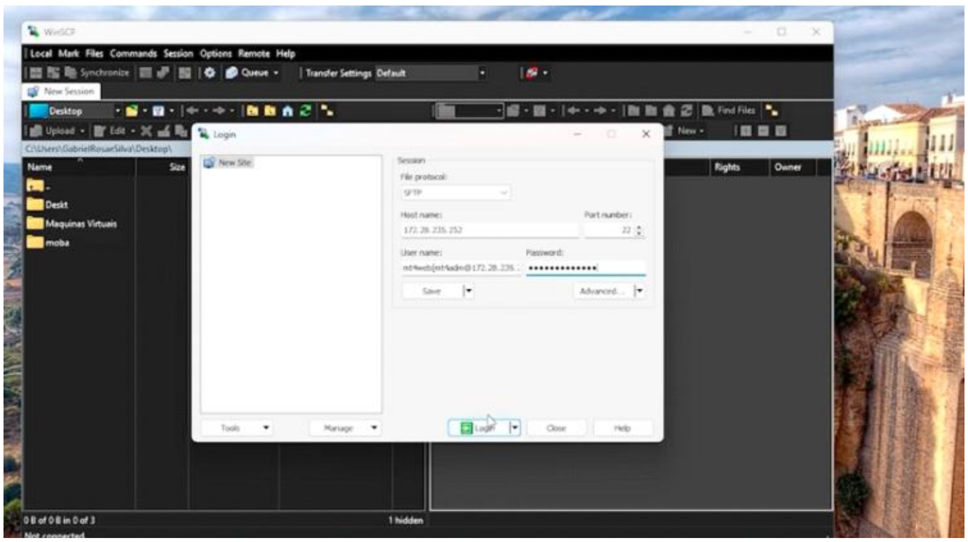

- Without login to server you can directly get a session by knowing server ip - onkey hop



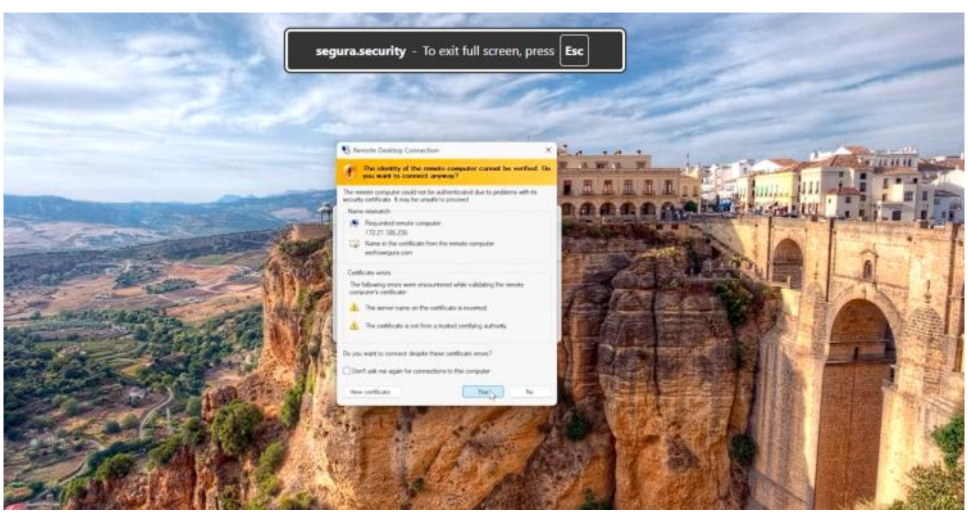

- You can use sftp for file transfer

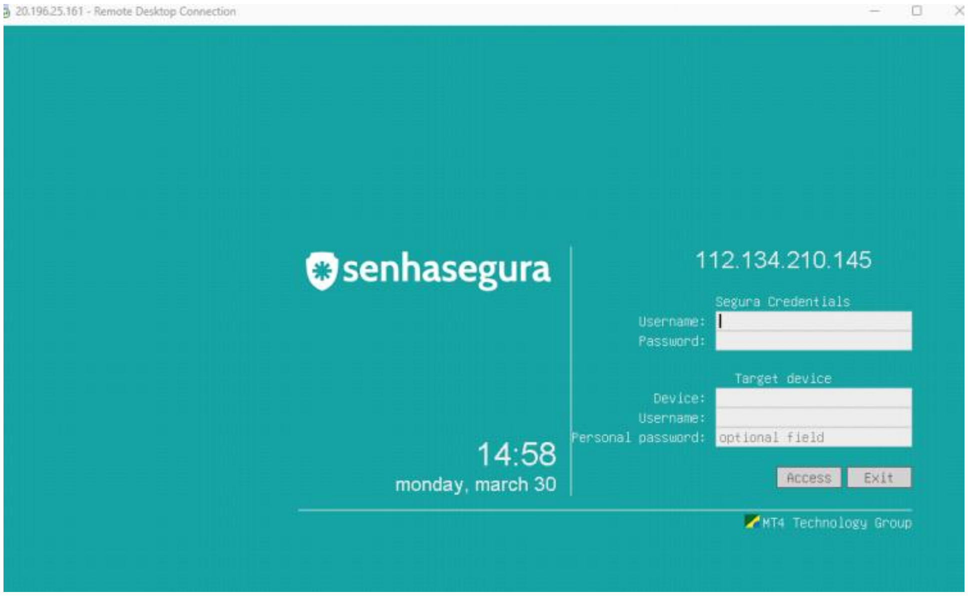



- Using winscp also you can transfer file

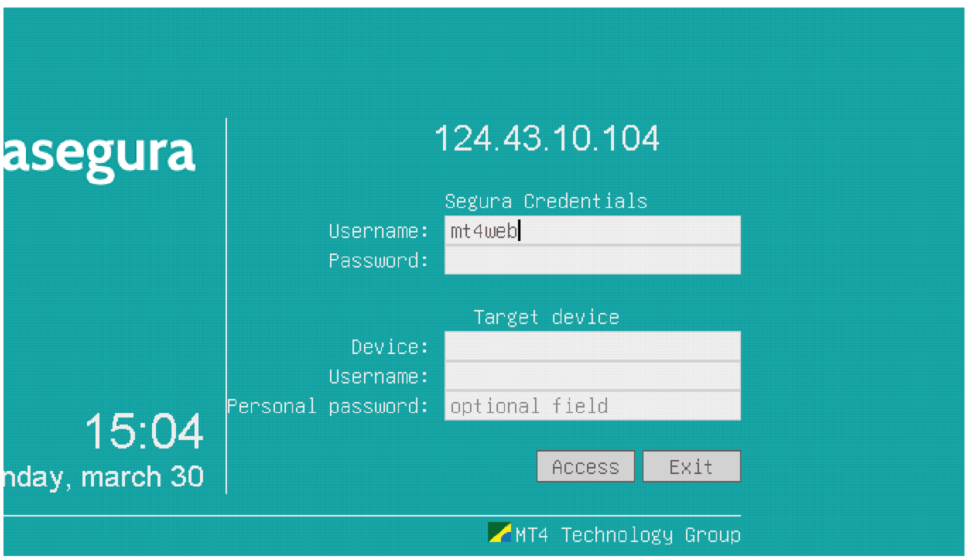


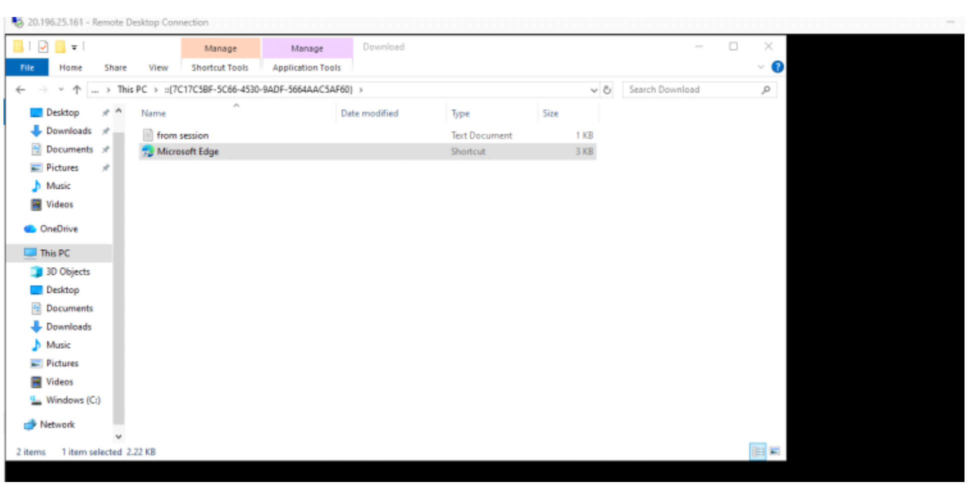

**You can initialize a session not only throught web instead xstream server mens getting access to proxy terminal  -- rdp**






By accessing RDP terminal we can eshtablish session from there



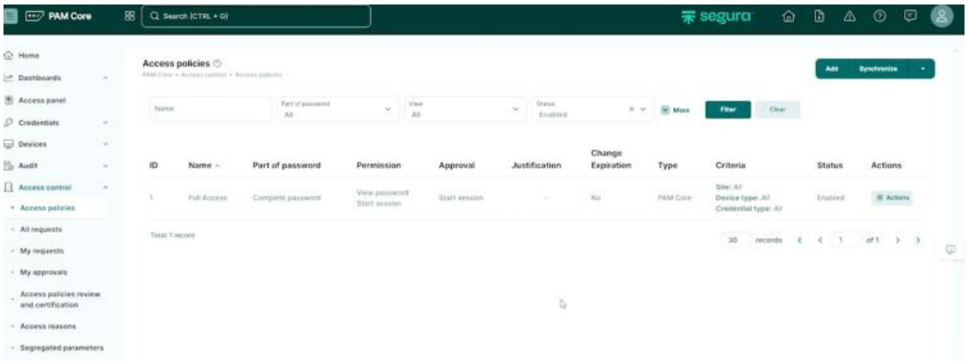

**Multihop also possible**


Remote Desktop Connection
Remote Desktop
Connection
General
Display
Local Resources
Experience
Advanced
Logon settings
Enter the name of the remote computer
Computer:
172.21.105.235
User name:
You will be asked for credentials when you connect.
Allow me to save credentials
Connection settings
Save the current connection settings to an RDP file or open a saved connection.
Save
Save As...
Open...
Hide Options
Connect
Help

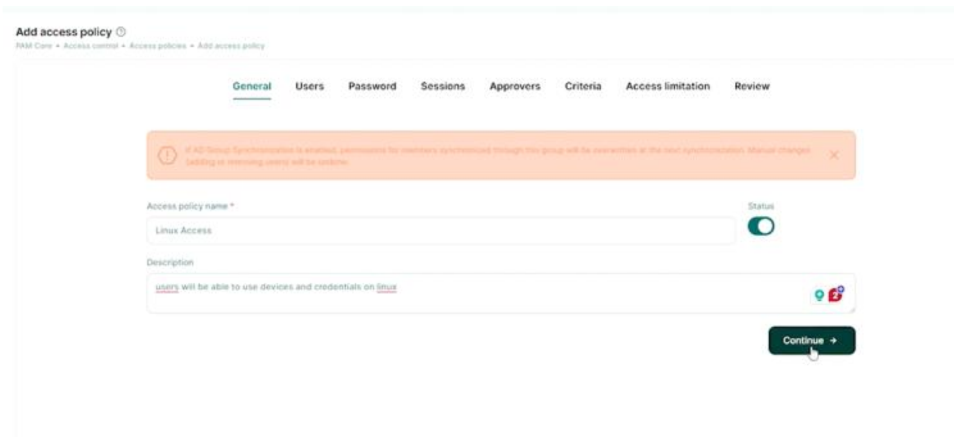
# Permissions - Access Policie

Wednesday, April 01, 2026 9:00 AM

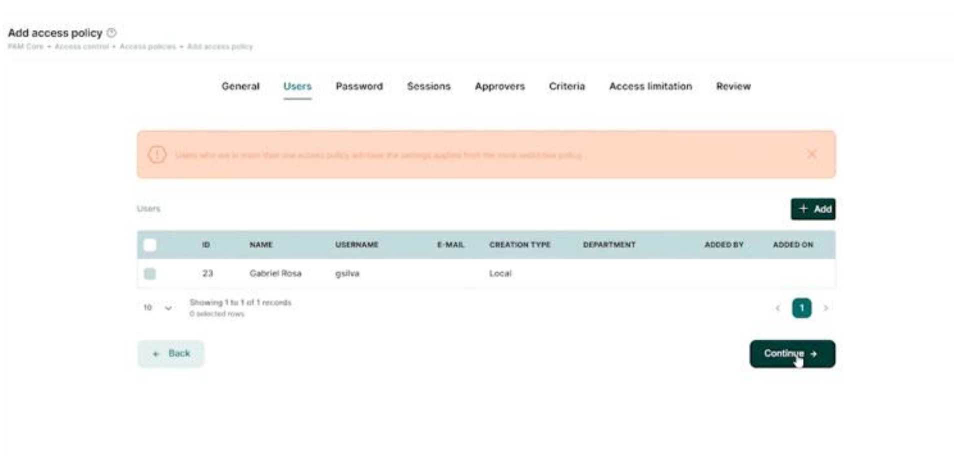
Access Policie : define what users can see in sensegura



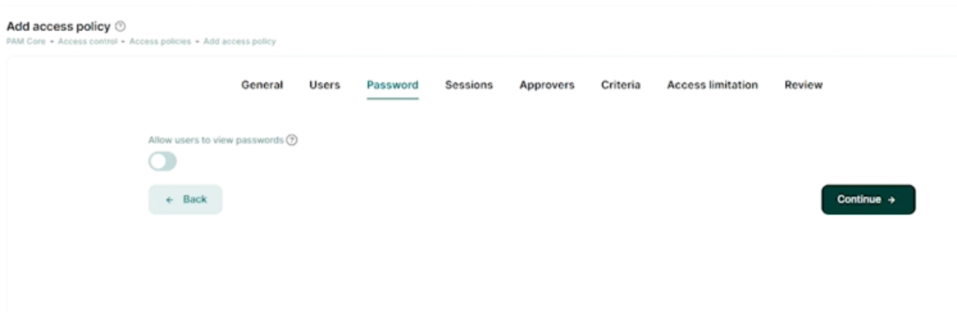

Info

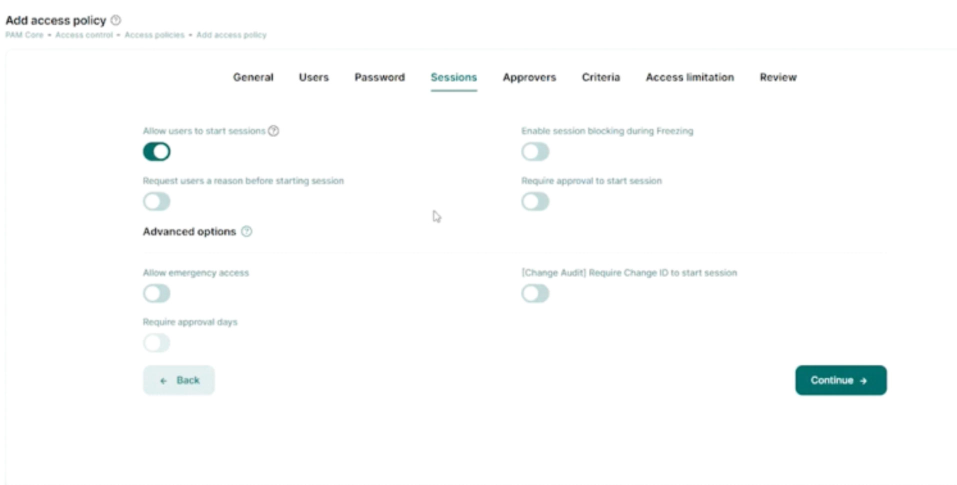


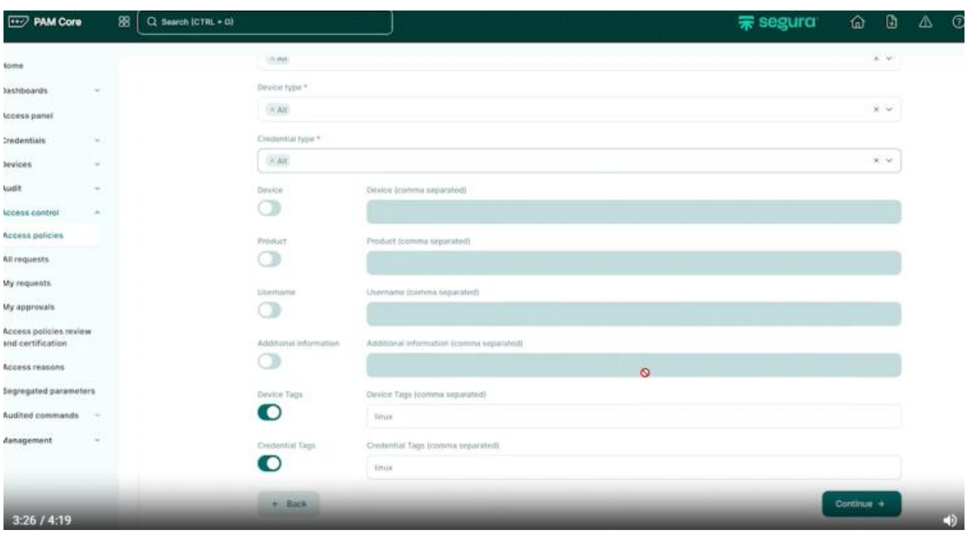

Add the user



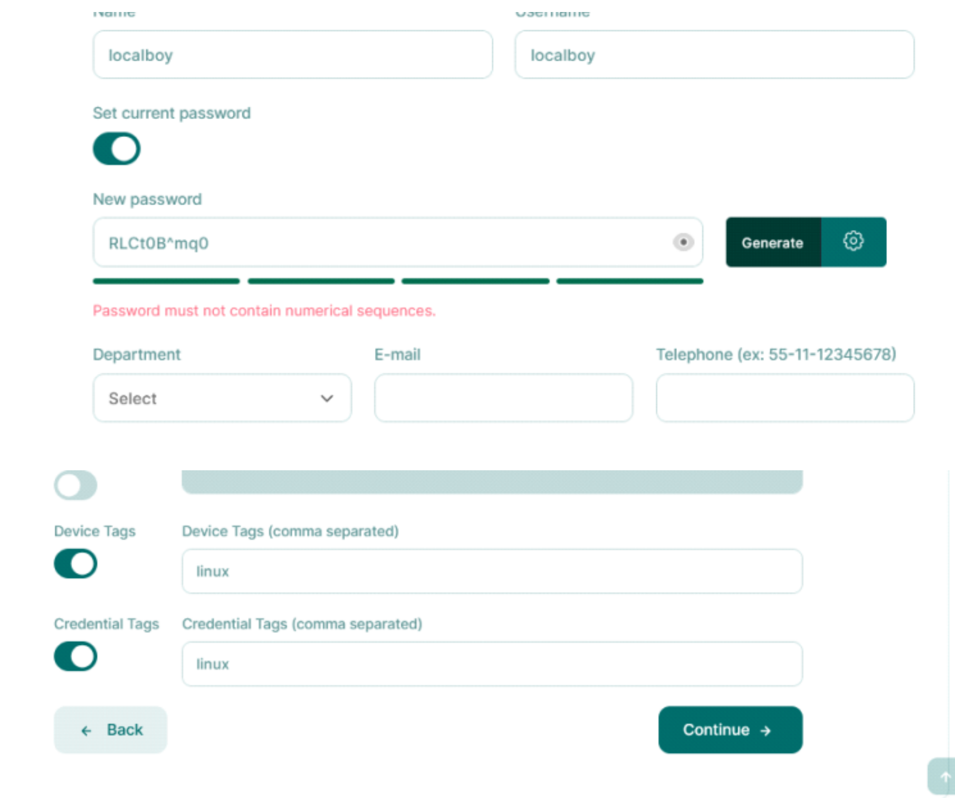

Aloow dissallow to view password



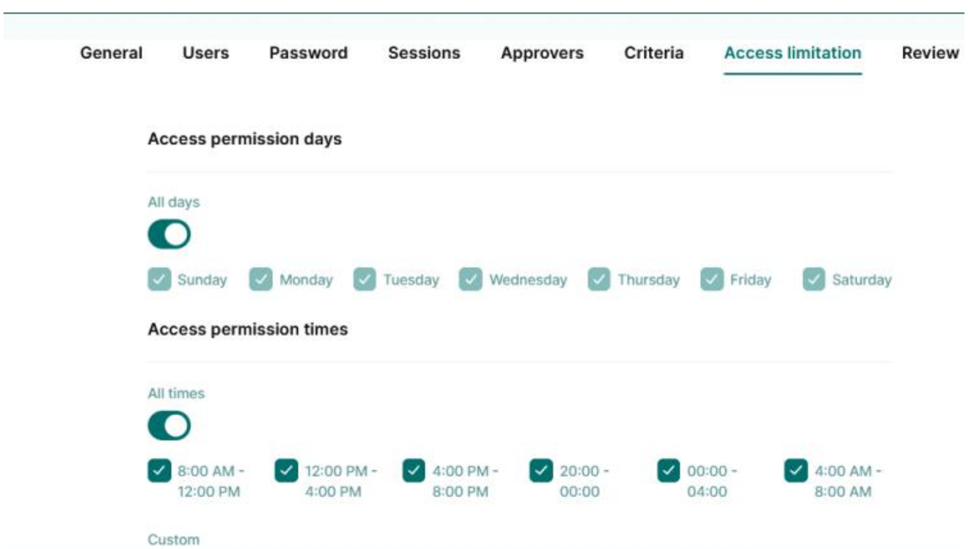



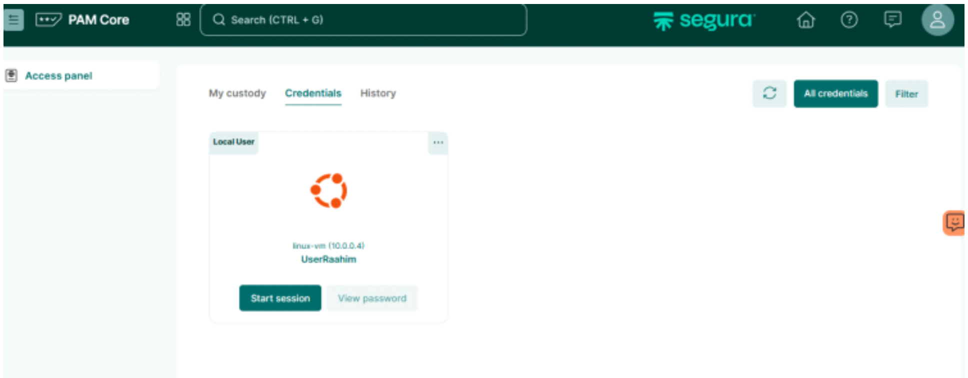

Create a user



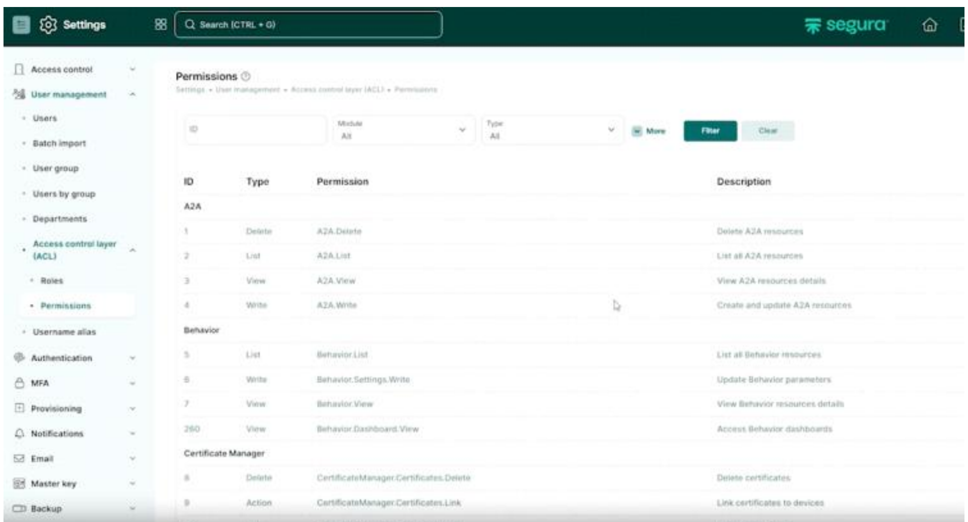

Limiting the accesses

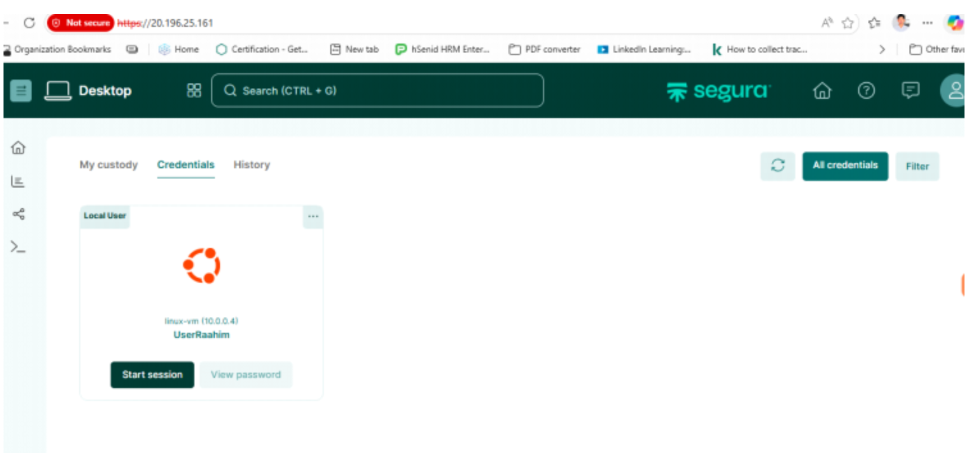



User end




Roles and permission

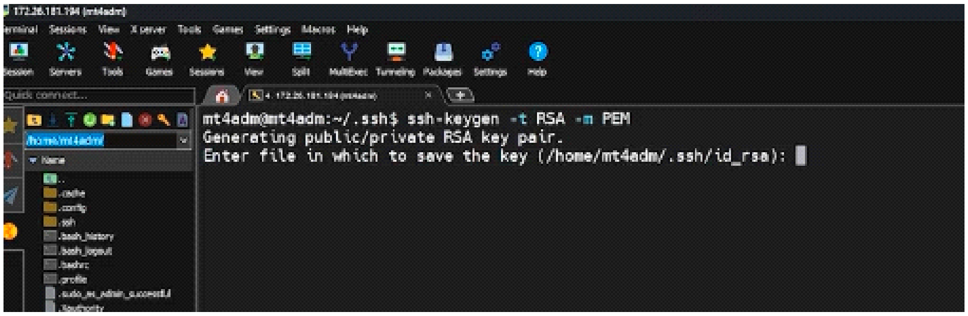


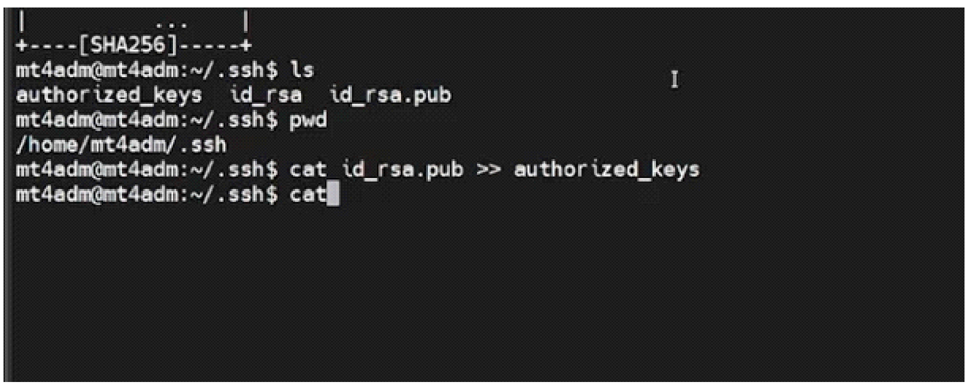
# SSH Key based login for linux

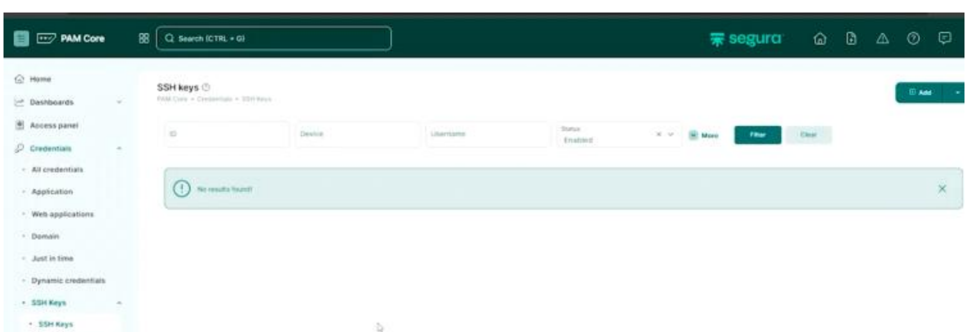
Tuesday, March 31, 2026 4:17 PM

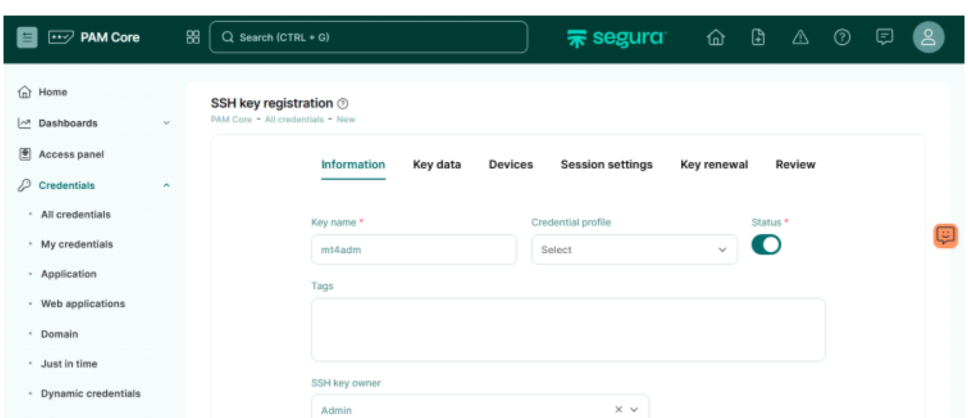
- Generate the key in the pam server



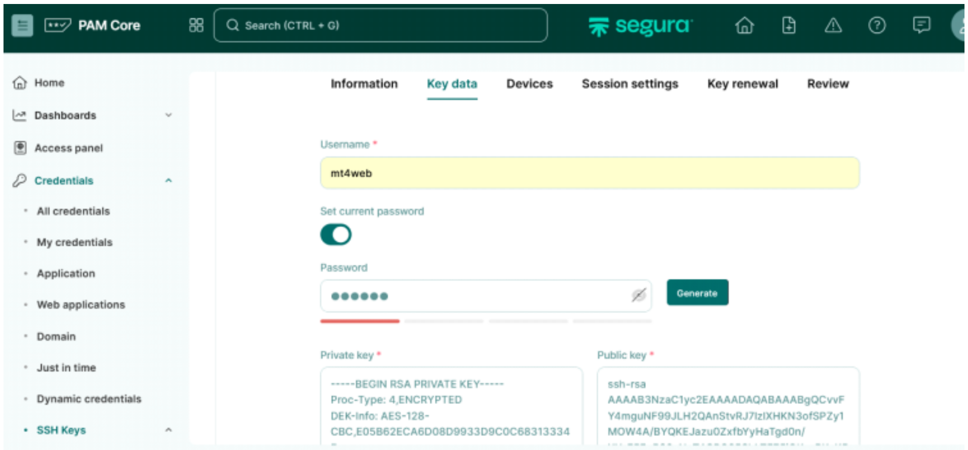

- Add he public key to the web interface

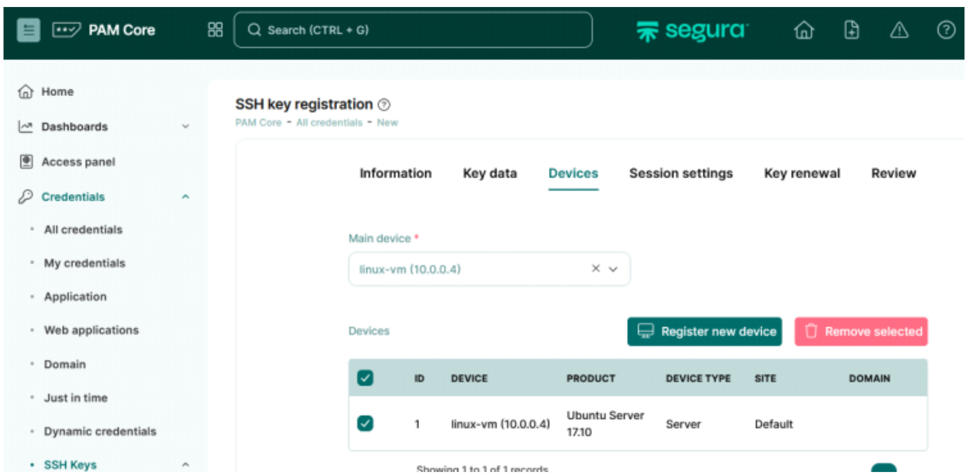


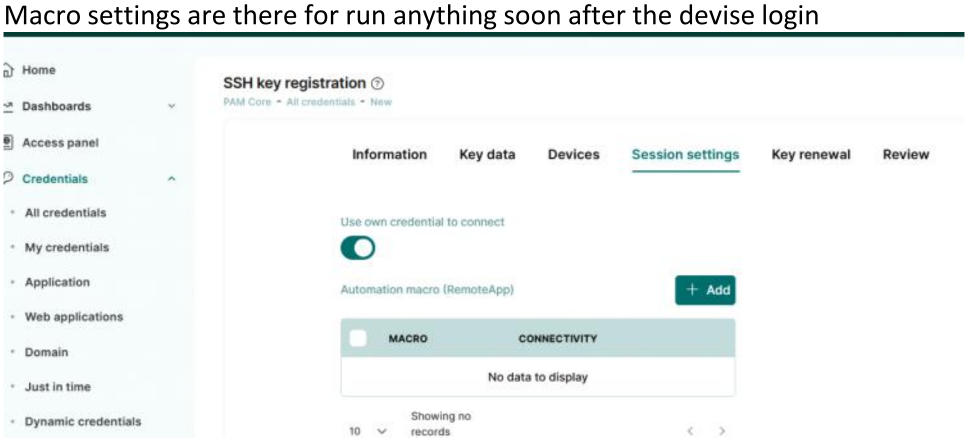



Key data

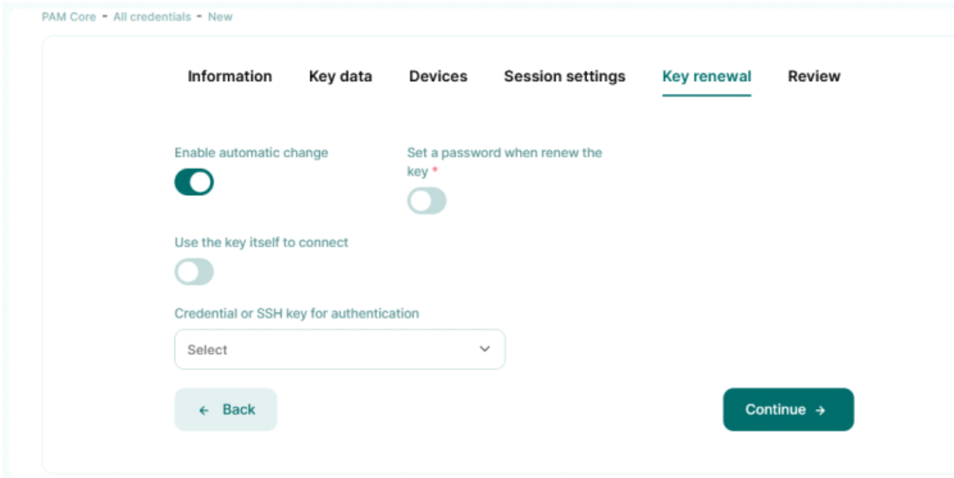


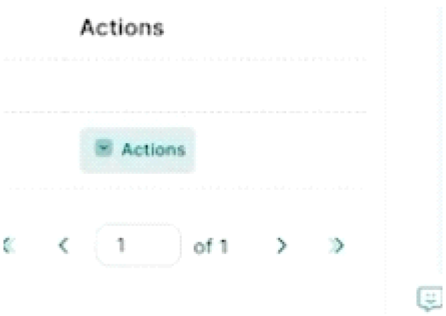




Macro settings are there for run anything soon after the devise login

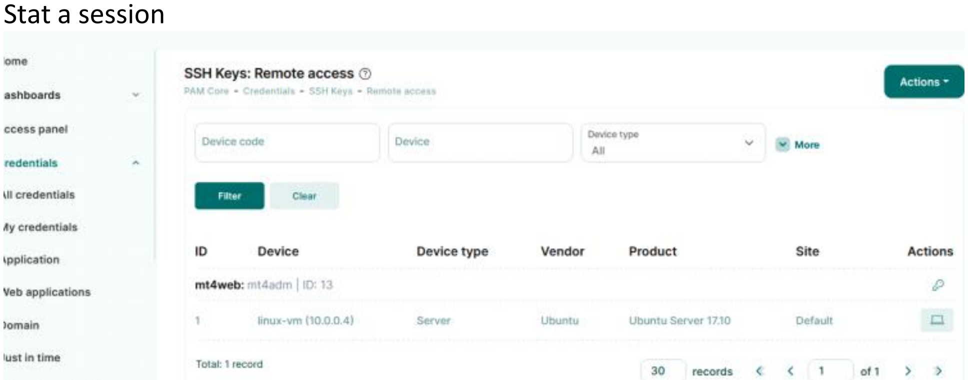



Key renewal




Stat a session



# Key gen

Wednesday, March 04, 2026 1:09 AM

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> ls ~/.ssh
PS C:\WINDOWS\System32> ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\RaahimMahmooth/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\RaahimMahmooth/.ssh/id_ed25519
Your public key has been saved in C:\Users\RaahimMahmooth/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:JsLC1KT/OQDeeyWckPXWxF13qLarP+3NYl9UxEi/HTo azuread\raahimmahmooth@ZLK256
The key's randomart image is:
+--[ED25519 256]--+
|    . .  ... oo++|
|   + o . o. . o+o|
|  + +   o .  . .o|
| + = o o    o . =|
|  + * = S  . E .o|
|   . = *    . .. |
|    . =      o  .|
|     . .    o +o.|
|           .o.+.o+|
+----[SHA256]-----+
```

```
|     . .    o +o.|
|           .o.+.o+|
+----[SHA256]-----+
PS C:\WINDOWS\System32> dir $env:USERPROFILE\.ssh


    Directory: C:\Users\RaahimMahmooth\.ssh


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          3/4/2026   1:08 AM            419 id_ed25519
-a----          3/4/2026   1:08 AM            112 id_ed25519.pub


PS C:\WINDOWS\System32> $env:USERPROFILE
C:\Users\RaahimMahmooth
PS C:\WINDOWS\System32>
```

```
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\WINDOWS\System32> type $env:USERPROFILE\.ssh\id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGeiNSlA7ZsHNMCVL91rjiFx5srX+skUHLMjGOwGNDoc azuread\raahimmahmooth@ZLK256
PS C:\WINDOWS\System32>
```

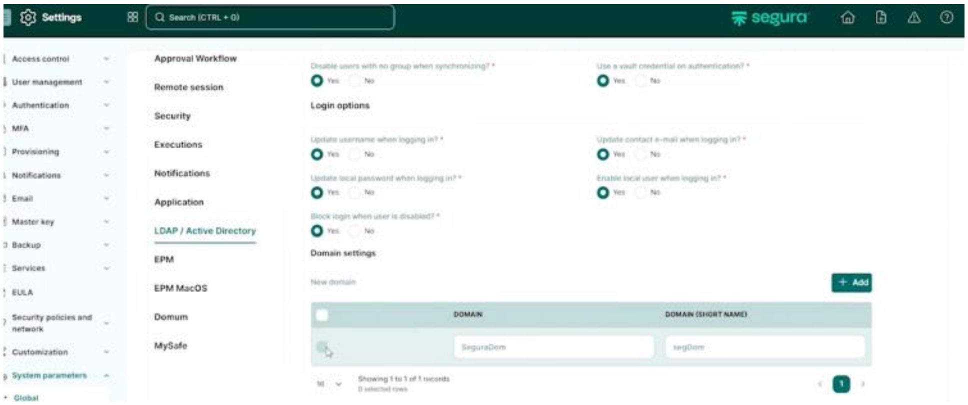
Ah, I get you now—thanks for clarifying. So, even though port 22 is open in the Azure NSG (Network Security Group), the VM image itself might have SSH disabled or misconfigured. Sometimes these marketplace images restrict SSH access for security purposes, or they might expect you to use a custom serial console setup.

In this case, I'd suggest using the Azure Serial Console. It's built into the Azure portal and doesn't rely on SSH. From there, you can log in, check the SSH configuration inside the VM, and correct any issues—like ensuring the SSH daemon is running or that the user is allowed to log in. Once you fix it from the serial console, you should be able to connect via SSH normally. Let me know if you need a hand with the steps to use the serial console!

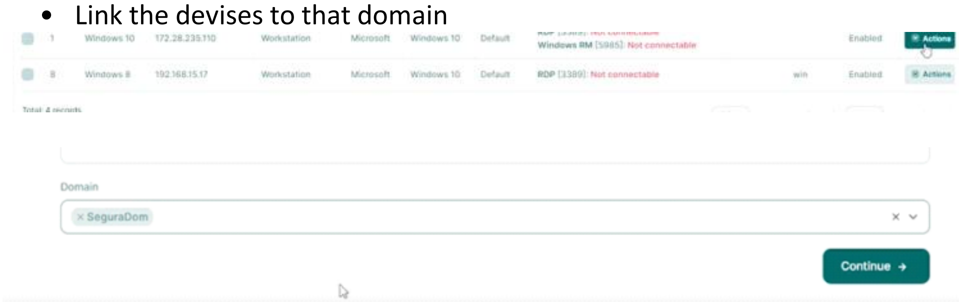
# Configuring Domain-Based Devices and Credentials

Wednesday, April 01, 2026 8:52 AM

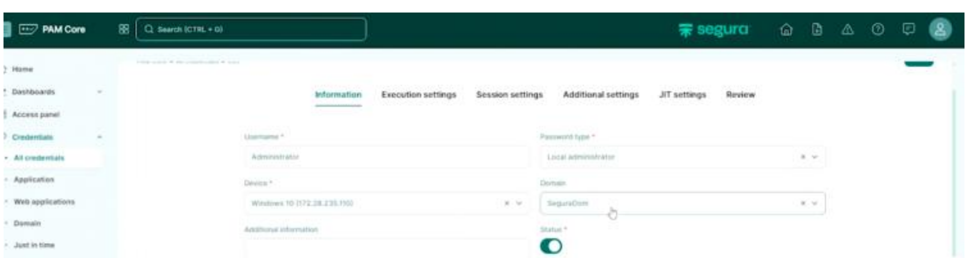
- This way our credentials can accesses any computer in that particular domain
- Add domain



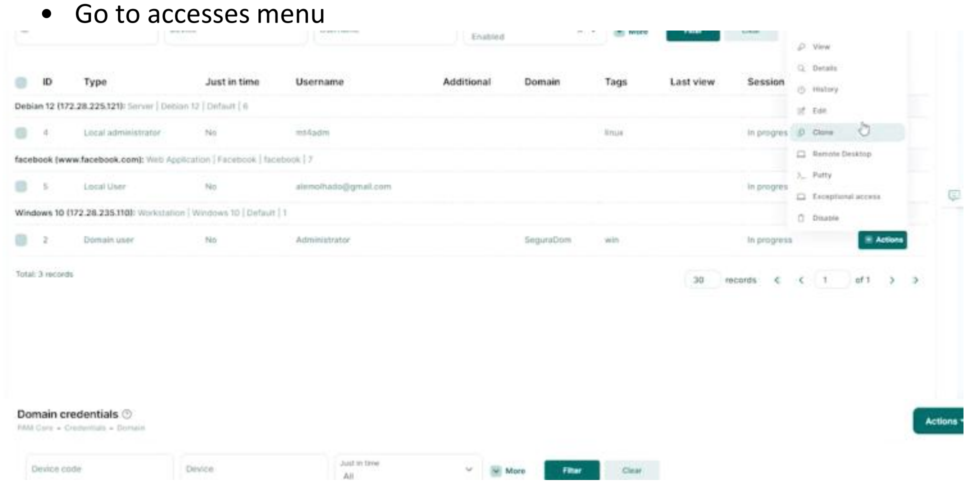

- Link the devises to that domain

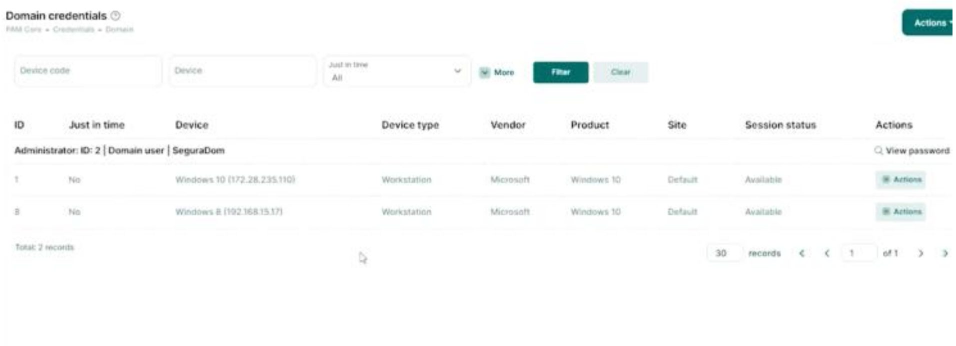



- Add credentials that domain




- Go to accesses menu




Domain credentials
PAM Core • Credentials • Domain
Actions
Device code
Device
Just in time
All
More
Filter
Clear
ID
Just in time
Device
Device type
Vendor
Product
Site
Session status
Actions
Administrator: ID: 2 | Domain user | SeguraDom
View password
1
No
Windows 10 (172.28.235.110)
Workstation
Microsoft
Windows 10
Default
Available
Actions
8
No
Windows 8 (192.168.15.17)
Workstation
Microsoft
Windows 10
Default
Available
Actions
Total: 2 records
30
records
1
of 1

# Proxy configuration

Thursday, April 16, 2026 11:40 AM

**proxy-rdp.service**

```
[Unit]
Description=proxy-rdp service
After=network-online.target docker.service proxy-core.service
Wants=network-online.target
Requires=docker.service

[Service]
TimeoutStartSec=60
Restart=always
RestartSec=30

ExecStartPre=/bin/sleep 60
ExecStartPre=-/usr/bin/docker stop proxy-rdp
ExecStartPre=-/usr/bin/docker rm proxy-rdp

ExecStart=/usr/bin/docker run --rm --name proxy-rdp \
-p 3390:3389 \
--env LANG="en_US.UTF-8" \
--env LANGUAGE="" \
--env LC_ALL="" \
-v /srv/cache/sessao:/srv/cache/sessao \
-v /etc/localtime:/etc/localtime:ro \
-v /etc/senhasegura/ssl/client_cert.crt:/etc/senhasegura/ssl/client_cert.crt:ro \
-v /etc/senhasegura/ssl/client_key.pem:/etc/senhasegura/ssl/client_key.pem:ro \
-v /etc/senhasegura/ssl/ca.crt:/etc/senhasegura/ssl/ca.crt \
-v /etc/senhasegura/proxy/proxy-rdp.conf:/etc/senhasegura/proxy/proxy-rdp.conf:ro \
-v /etc/senhasegura/ssl/proxy-rdp.crt:/etc/senhasegura/ssl/proxy-rdp.crt:ro \
-v /etc/senhasegura/ssl/proxy-rdp.key:/etc/senhasegura/ssl/proxy-rdp.key:ro \
-v /dev/senhasegura-logs/proxy_ng:/dev/senhasegura-logs/proxy_ng \
--log-driver syslog \
--log-opt syslog-address=unixgram:///dev/senhasegura-logs/proxy_ng/rdp \
--log-opt tag="{{.Name}}" \
proxy-rdp:4.0.1-6
```

ExecStop=/usr/bin/docker rm -f proxy-rdp

[Install]

WantedBy=multi-user.target

**proxy-terminal.service**

```
[Unit]
Description=proxy-terminal service
After=network-online.target docker.service proxy-core.service
Wants=network-online.target
Requires=docker.service

[Service]
TimeoutStartSec=180
Restart=always
RestartSec=30

ExecStartPre=-/usr/bin/docker stop proxy-terminal
ExecStartPre=-/usr/bin/docker rm proxy-terminal
ExecStartPre=/bin/sleep 20

ExecStart=/usr/bin/docker run --rm --name proxy-terminal \
-p 22:22 \
--env LANG="en_US.UTF-8" \
--env LANGUAGE="" \
--env LC_ALL="" \
-v /srv/cache/sessao:/srv/cache/sessao \
-v /etc/localtime:/etc/localtime:ro \
-v /etc/senhasegura/ssl/client_cert.crt:/etc/senhasegura/ssl/client_cert.crt:ro \
-v /etc/senhasegura/ssl/client_key.pem:/etc/senhasegura/ssl/client_key.pem:ro \
-v /etc/senhasegura/ssl/ca.crt:/etc/senhasegura/ssl/ca.crt \
-v /dev/senhasegura-logs/proxy_ng:/dev/senhasegura-logs/proxy_ng \
--log-driver syslog \
--log-opt syslog-address=unixgram:///dev/senhasegura-logs/proxy_ng/terminal \
--log-opt tag="{{.Name}}" \
proxy-terminal:4.0.1-1
```

**ExecStop=/usr/bin/docker rm -f proxy-terminal**

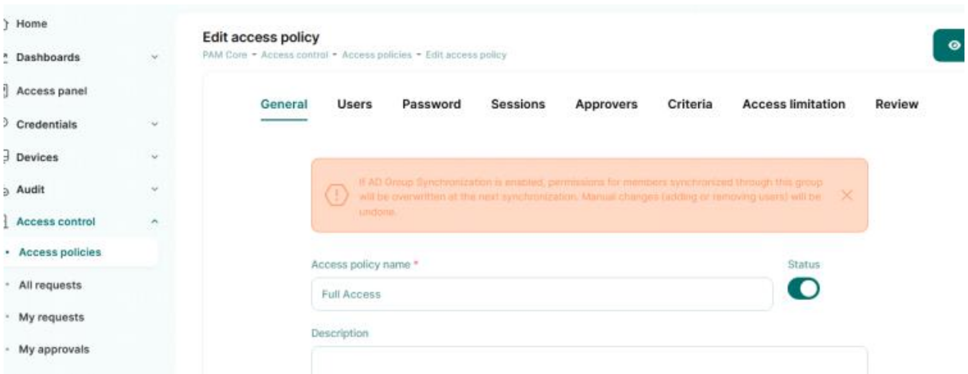
**[Install]**

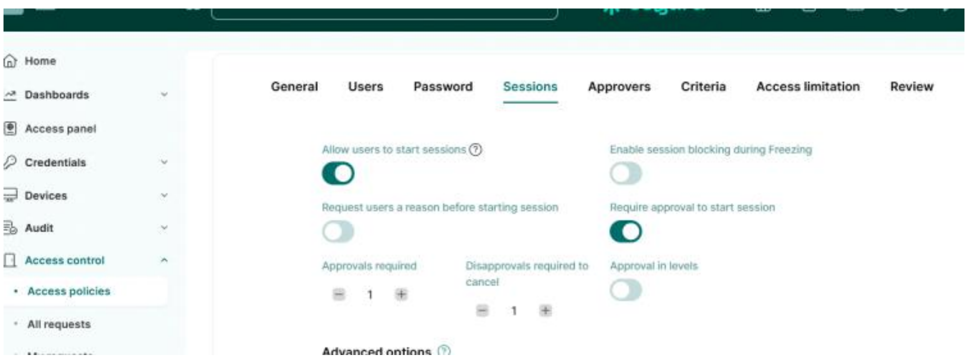
**WantedBy=multi-user.target**

# Approval workflow

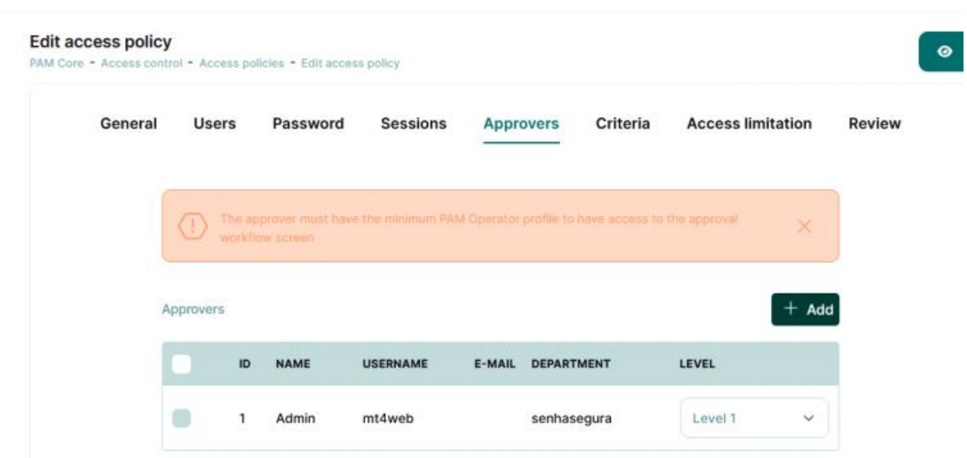
Wednesday, April 01, 2026 9:56 AM

- Edit access policy

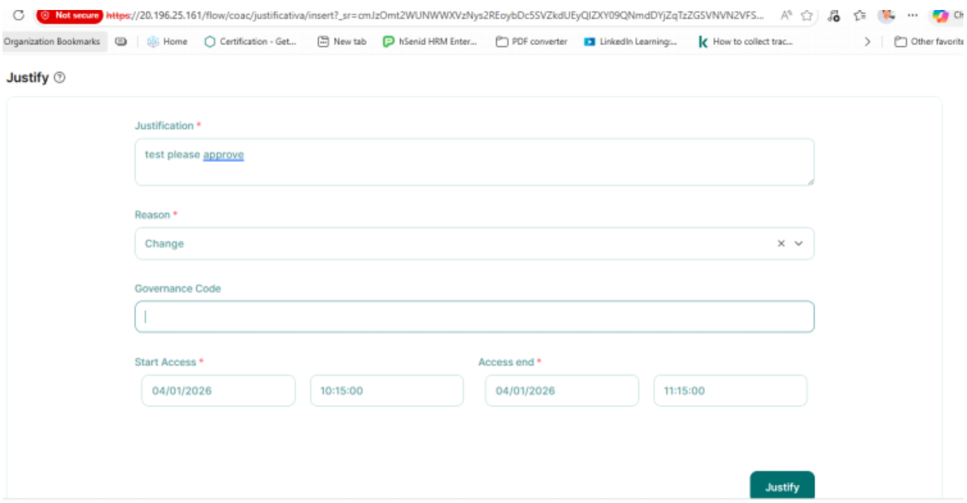


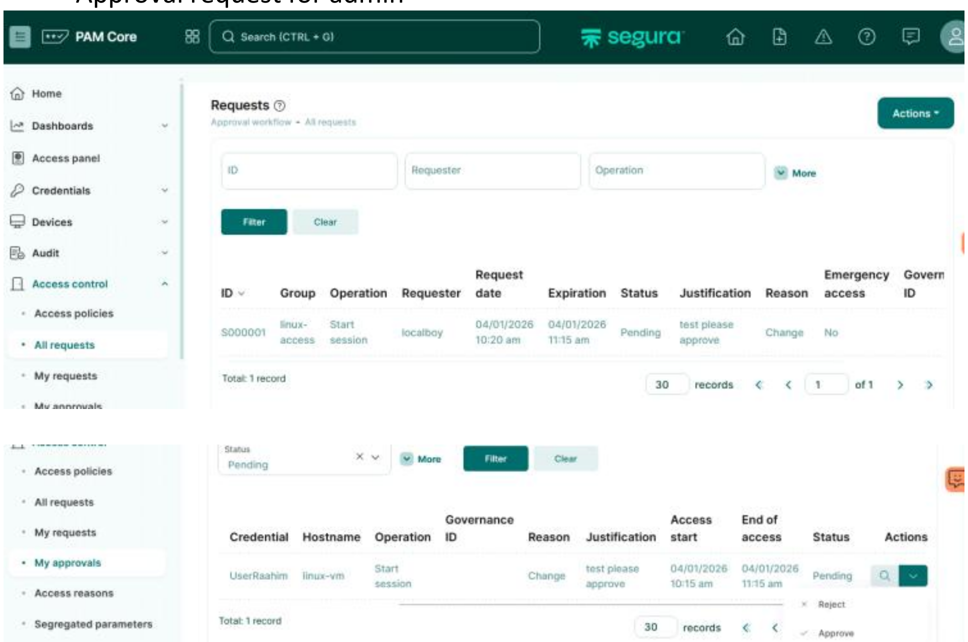

- Define who is the approver



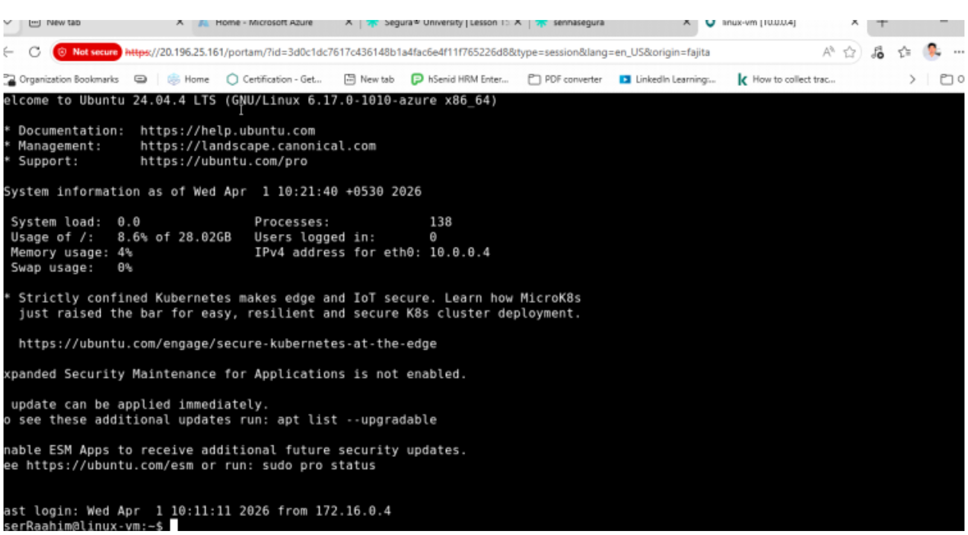

- Justify your reason when your starting a session



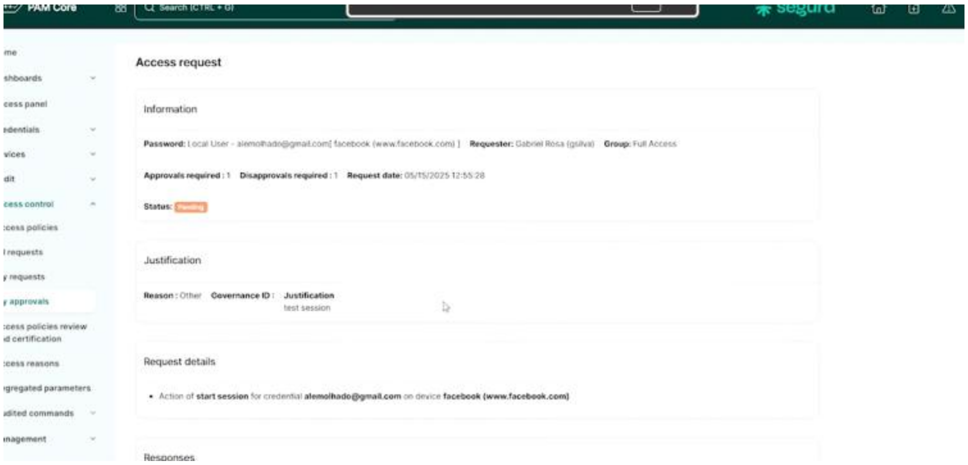

- Approval request for admin




- Local user able to accesse



You can view the nature of the request history



# troubleshooting

Thursday, March 05, 2026 11:01 AM

Issue : The PAM system successfully authenticated the user and initiated a session, but **failed to establish the backend connection to the target machine via Terminal Proxy / RDP Proxy**, resulting in session failure.

Sessions were stuck after creation due to proxy service failure, even though authentication and connectivity were working.

**Terminal Proxy = the middleman that creates the session between you and the server**

You (Browser)
↓
Senhasegura Web
↓
Terminal Proxy (container)
↓
Target VM (SSH / RDP)

**Terminal container = a Docker container that runs the Terminal Proxy service**


1. PAM calls → Terminal container
2. Container opens SSH connection to target
3. Container streams terminal back to browser

This container behaves like a **mini SSH server**

That's why:

- It needs a **port (22 by default)**
- It caused your issue:

Terminal container is the Docker container that runs the SSH proxy service used by PAM to create sessions.

**Look for logs like:**

- `secure`
- `messages`
- `Auth.log`

```
user.log
```

```
Mar 19 06:58:22 segura senhasegura[707530]: Mar 19 06:58:22 segura CEF:0|MT4
|senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4
spid=707530 src=127.0.0.1 suid=11 sname=Nss Orbini [segura]
suser=_a2a_nss_orbini_segura_ spriv=User tenant=senhasegura msg=User not
```

found requestMethod=POST act=Authentication error cs1Label=Plugin cs1 =Terminal Proxy cs2Label=Error code cs2=-1 cs5Label=Country code cs5=- cs6Label=Country cs6=-

Mar 19 06:58:45 segura senhasegura[710281]: Mar 19 06:58:45 segura CEF:0|MT4 |senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4 spid=710281 src=127.0.0.1 suid=11 sname=Nss Orbini [segura] suser=_a2a_nss_orbini_segura_ spriv=User tenant=senhasegura msg=User not found requestMethod=POST act=Authentication error cs1Label=Plugin cs1 =Terminal Proxy cs2Label=Error code cs2=-1 cs5Label=Country code cs5=- cs6Label=Country cs6=-

Mar 19 06:59:12 segura senhasegura[712860]: Mar 19 06:59:12 segura CEF:0|MT4 |senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4 spid=712860 src=127.0.0.1 suid=11 sname=Nss Orbini [segura] suser=_a2a_nss_orbini_segura_ spriv=User tenant=senhasegura msg=User not found requestMethod=POST act=Authentication error cs1Label=Plugin cs1 =Terminal Proxy cs2Label=Error code cs2=-1 cs5Label=Country code cs5=- cs6Label=Country cs6=-

messages

Mar 19 07:52:43 segura senhasegura[987174]: Mar 19 07:52:43 segura CEF:0|MT4 |senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4 spid=987174 src=124.43.16.212 suid=14 sname=Raahim suser=raahimmahmooth spriv=User tenant=senhasegura msg=Authenticated successfully requestMethod=POST act=Authentication successfully cs1Label=Plugin cs1 =Auth_Adapter_Local cs3Label=Redirect cs3=Redirected to enter the access token cs5Label=Country code cs5=LK cs6Label=Location cs6=

Mar 19 07:55:07 segura senhasegura[1000241]: Mar 19 07:55:07 segura CEF:0|MT4 |senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4 spid=1000241 src=124.43.16.212 suid=14 sname=Raahim suser=raahimmahmooth spriv=Administrator tenant=senhasegura msg=Token validated! requestMethod=POST act=Multi-factor authentication cs1Label=Trigger cs1=Token TOTP

Mar 19 07:55:07 segura senhasegura[1000241]: Mar 19 07:55:07 segura CEF:0|MT4 |senhasegura|4.0.12-2|1695.001|Orbini - Authentication|5| dvc=172.16.0.4 spid=1000241 src=124.43.16.212 suid=14 sname=Raahim suser=raahimmahmooth spriv=Administrator tenant=senhasegura msg=Token validated! requestMethod=POST act=Multi-factor authentication cs1Label=Plugin cs1=Google cs5Label=Country code cs5=LK cs6Label=Location cs6=

Mar 19 07:55:30 segura senhasegura[1001851]: Mar 19 07:55:30 segura CEF:0|MT4 |senhasegura|4.0.12-2|16|Session started|5| dvc=172.16.0.4 spid=1001851 src= 124.43.16.212 suid=14 sname=Raahim suser=raahimmahmooth spriv=User tenant=senhasegura msg=A new session was started by the user Raahim (raahimmahmooth) using the credential Local User - UserRaahim[ linux-vm (10.0.0.4) ] on the device linux-vm (10.0.0.4) requestMethod=GET act=Session started dst=10.0.0.4 dpt=22 duid=7 proto=ssh duser=UserRaahim sourceServiceName=Web Proxy cs1Label=Session ID cs1= 4a1a5cb5d38d2981c1b36442d2a6aa86b6cc731a dhost=linux-vm

Mar 19 07:55:30 segura senhasegura[1002169]: Coss_Iso_Plugin:Coss_Iso_Plugin::getSessao: Session is already started

### Key lines checked

- `UsePAM yes` → ✓ **Required for authentication**
- `AllowTcpForwarding yes` → ✓ fine
- `PermitTTY yes` → ✓ required for terminal sessions (important for PAM)
- `X11Forwarding yes` → ✓ optional (not an issue)
- `UseDNS no` → ✓ good (faster SSH)
- `MaxStartups 10:30:100` → ✓ fine

**Watched real time logs**

- `sudo journalctl -u guacd –f – watch guacd logs`
- `sudo tail -f /var/log/messages`

sudo systemctl restart guacd - restart (It is the engine that actually connects to remote machines)

sudo systemctl restart nginx | apache

sudo journalctl -u guacd -f - while starting the session to see whats happening in backend

**--can be like this ---**

proxy-terminal.service: Failed with result 'exit-code'

proxy-rdp.service: Failed with result 'exit-code'

**---status check of both proxy---**

sudo systemctl status proxy-terminal.service

sudo systemctl status proxy-rdp.service

**---Check Docker containers---**

docker ps -a

----continer logs----

docker logs <container_id>

**--- restart the services & docker---**

sudo systemctl restart docker

sudo systemctl restart proxy-terminal.service

sudo systemctl restart proxy-rdp.service

**----- Check required directory-----**

Run:

ls -ld /srv/cache/sessao

❌ If NOT exists → CREATE:

sudo mkdir -p /srv/cache/sessao

sudo chmod -R 777 /srv/cache

**----Check disk space---**

df -h

**---starting and getting error ---**

journalctl -u proxy-terminal.service -n 50 --no-pager

journalctl -u proxy-rdp.service -n 50 --no-pager

**--- — IMMEDIATELY check----**

docker ps -a | grep proxy

**-----☑ EXPECTED RESULT-----**

You should now see:

proxy-terminal   Up ...

proxy-rdp        Up ...

**---Check Docker image exists ---**

docker images | grep terminal

**----prints the error---**

sudo docker run --rm --net host \
-v /srv/cache/sessao:/srv/cache/sessao \
proxy-terminal:4.0.1-1

**---binding a port to use by doker---**

sudo nano /lib/systemd/system/proxy-terminal.service
edit this line::ExecStart=/usr/bin/docker run --rm --name proxy-terminal \
-p 2222:22 \

**---relod the systemd---**
sudo systemctl daemon-reexec
sudo systemctl daemon-reload

**---see what port the sshd uses---**
sudo nano /etc/ssh/sshd_config

# commands

Wednesday, March 04, 2026 2:44 PM

- mt4adm -console default password and username
- 
- sudo su
- orbit hostname <change- me>
- orbit network - set static ip,netmask, gateway
- orbit lacale timezoene
- orbit shutdown -r
- 
- check is the console can access repos

- telnet deb.senhasegura.com 443
- telnet repo.senhasegura.com 443

- updating -online update and offline update is possible

- wget https://downloads.senhasegura.io/d/other/aptfix
- sudo chmod +x aptfix
- sudo .aptfix

- apt update
- apt-get install orbit-cli

- orbit update
- orbit update --application - application only
- orbit version

- Ssh –p <> username@ip

- Orbit app status
- ssh -p 59022 mt4adm@20.196.25.161 - VEElmq@20031122
- ssh UserRaahim@20.197.45.83 - VEElmq@20031122
- G#JEfe_13o

**--can be like this ---**

proxy-terminal.service: Failed with result 'exit-code'

proxy-rdp.service: Failed with result 'exit-code'

**---status check of both proxy---**

sudo systemctl status proxy-terminal.service

sudo systemctl status proxy-rdp.service

**---Check Docker containers---**

docker ps -a

----continer logs----

docker logs <container_id>

**--- restart the services & docker---**

sudo systemctl restart docker

sudo systemctl restart proxy-terminal.service

sudo systemctl restart proxy-rdp.service

**----- Check required directory-----**

Run:

ls -ld /srv/cache/sessao

❌ If NOT exists → CREATE:

sudo mkdir -p /srv/cache/sessao

sudo chmod -R 777 /srv/cache

**----Check disk space---**

df -h

**---starting and getting error ---**

journalctl -u proxy-terminal.service -n 50 --no-pager

journalctl -u proxy-rdp.service -n 50 --no-pager

**--- — IMMEDIATELY check----**

docker ps -a | grep proxy

**-----☑ EXPECTED RESULT-----**

You should now see:

proxy-terminal   Up ...

proxy-rdp        Up ...

**---Check Docker image exists ---**

docker images | grep terminal

**----prints the error---**

sudo docker run --rm --net host \

-v /srv/cache/sessao:/srv/cache/sessao \

proxy-terminal:4.0.1-1

**---binding a port to use by doker---**

sudo nano /lib/systemd/system/proxy-terminal.service

edit this line::ExecStart=/usr/bin/docker run --rm --name proxy-terminal \

-p 2222:22 \

**---relod the systemd---**

sudo systemctl daemon-reexec

sudo systemctl daemon-reload

**---see what port the sshd uses---**

sudo nano /etc/ssh/sshd_config

**Ssh**

```
sudo nano /etc/ssh/sshd_config
```

**Make sure these are enabled:**

```
PasswordAuthentication yes
PermitRootLogin yes   (optional)
```

**Then restart SSH:**

```
sudo systemctl restart ssh
```

## Logs

Look for logs like:

- secure
- messages
- auth.log

- sudo journalctl -u guacd –f – watch guacd logs
- sudo tail -f /var/log/messages

- docker images | grep terminal – checked image exiss or not